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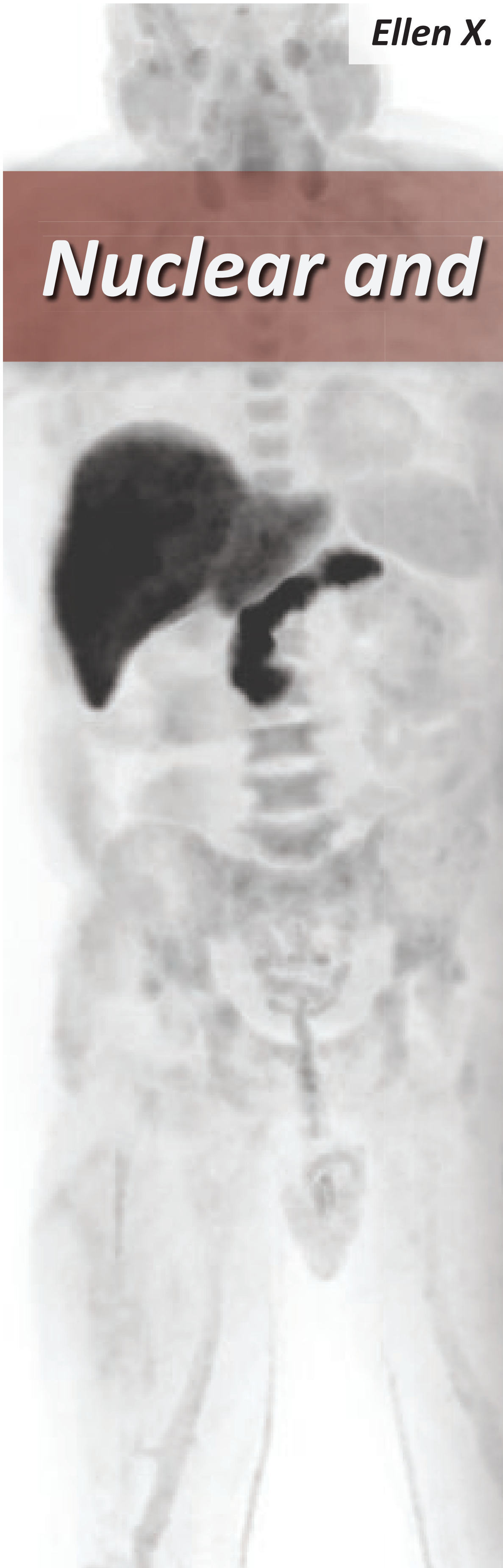
Nuclear and Molecular Imaging

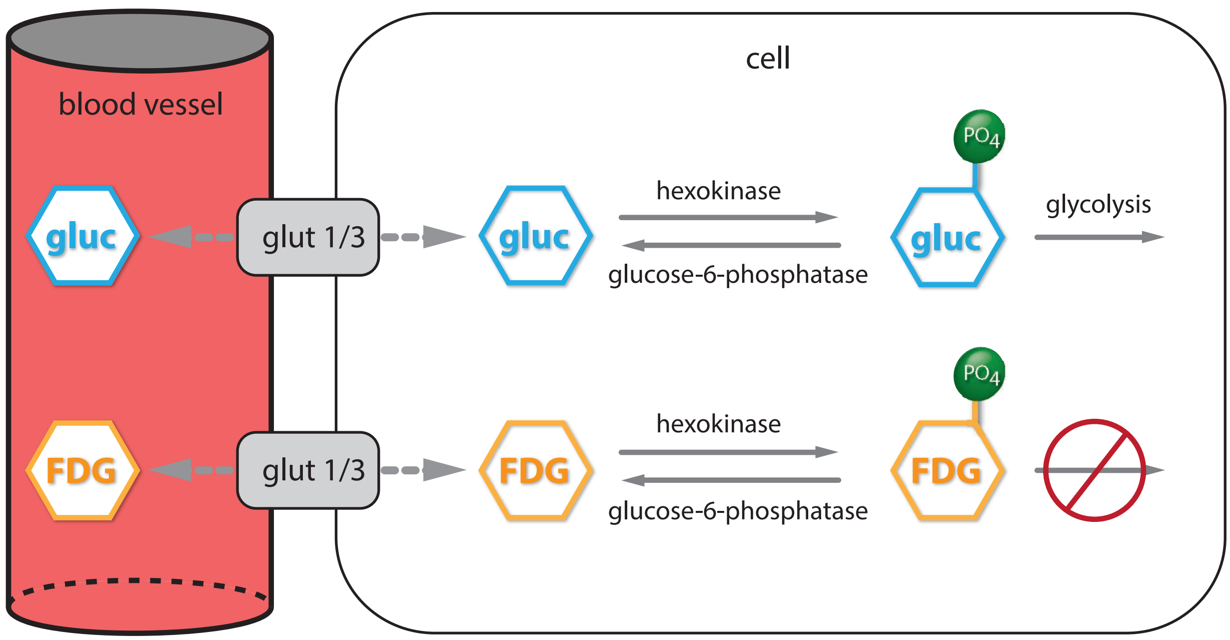


Basic positron emission tomography (PET) physics

- A positron emission tomography (PET) radiotracer emits a positron that travels a small local distance within the tissue. After meeting an electron, both the positron and electron annihilate and create two 511 keV photons travelling nearly 180 degrees apart.
- The two high-energy photons travelling in opposite directions are simultaneously detected by a circular ring of crystals, which can determine that the two photons arrived coincidentally.

Technical considerations of F-18 FDG PET/CT

Radiotracer: Fluorine-18 FDG



- Fluorine-18 (F-18) is a cyclotron-produced positron emitter, half-life 110 minutes.
- F-18-fluorodeoxyglucose (F-18 FDG) is a glucose analog that competes with glucose for transport into cells by the GLUT 1 and 3 transporters. After becoming phosphorylated by hexokinase, FDG-phosphate cannot undergo glycolysis and is effectively trapped within cells.

Radiotracer uptake quantification

- The **standardized uptake value (SUV)** roughly quantifies FDG uptake.
- SUV is proportional to (ROI activity * body weight) / administered activity.
- SUV of a region of interest can vary significantly depending on multiple factors including the specific equipment used, time elapsed after FDG administration, the presence of any tracer extravasation, blood glucose and insulin levels at time of injection, etc.
- An SUV value above 2.5 traditionally has been suggestive of malignancy. However, malignancy can never be definitively diagnosed or excluded using SUV as the sole criterion.
- Size threshold for lesion detection on FDG PET is 6 mm. For small lesions, the average SUV may be falsely low due to partial volume averaging; the maximum SUV is a better measure.
- An alternative approach to calculating SUV values is visual or quantitative assessment of FDG uptake relative to background activity in normal tissues (e.g., liver or cerebellum) or the mediastinal blood pool.

CT correlation

- Most modern PET studies are performed together with CT as a PET/CT. One important exception is in the pediatric population, where PET may be performed in isolation to reduce the risk of radiation exposure from CT.
- The CT exam is often performed with a lower dose than a diagnostic-quality CT. The CT protocol varies by institution (e.g., whether intravenous contrast is administered).
- The CT is used for anatomic localization and attenuation correction.
- Very dense retained oral contrast or dense metallic objects (such as joint prostheses) may cause artifactual FDG uptake due to miscalculation of attenuation correction.

Patient preparation

- The uptake of FDG in both normal and pathological tissues is dependent on the serum glucose and insulin levels.
- Elevated insulin levels will cause increased muscle uptake and decrease the sensitivity for detecting mildly FDG-avid lesions.
- Patients should be NPO for at least 4 hours to allow insulin to reach a basal level.

  In type 1 diabetes, imaging should be performed in the morning after an overnight fast. Insulin is not recommended.

  In type 2 diabetes, regular insulin should be held 2–4 hours prior to the study. Short-acting insulin can be used but is not recommended. There is no need to hold long-acting insulin and oral hypoglycemic medications such as Metformin, though the latter can increase bowel uptake.
- Blood glucose should be below 200 mg/dL, preferably below 150.

  Elevated serum glucose will result in decreased FDG uptake by tumors and the brain.
- The typical administered dose of F-18 FDG is 10–25 mCi. After intravenous injection, the patient should rest in a quiet room for 60 minutes before imaging.

  If the patient talks, the vocal cords may show FDG uptake.

  If the patient walks, the muscles may show FDG uptake.

Normal FDG distribution

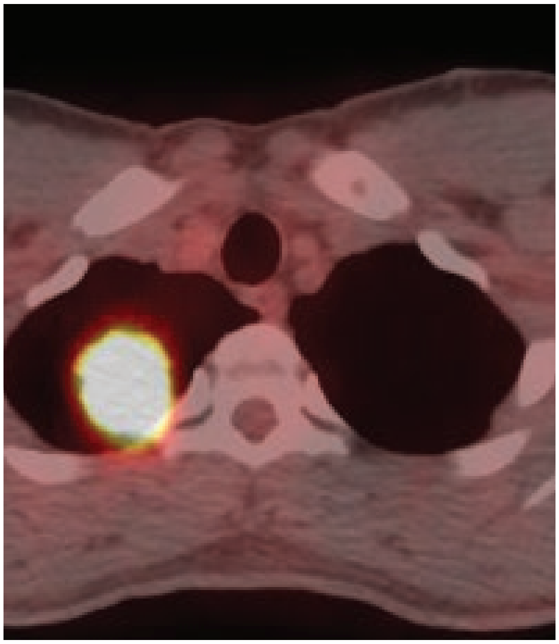
- *Brain:* The brain has intense FDG uptake. Brains love sugar!

  Despite intense uptake, with appropriate windowing, excellent detail of the cortex, basal ganglia, and cerebellum can be seen.
- *Renal, ureters, and bladder:* FDG is concentrated in the urine, with very intense activity in the renal collecting system, ureters, and bladder. As a result, bladder is the **critical organ** (the body part most susceptible to radiation damage from a given isotope). Good hydration is recommended for clearance of radioactivity.
- *Salivary glands, tonsils, thyroid:* Mild to moderate symmetric uptake.
- *Liver:* Moderate, largely homogeneous uptake.
- *Bowel:* Diffuse mild to moderate uptake is normal. Focal uptake within the bowel, however, should be regarded with suspicion.

  Metformin can increase colonic, and to a lesser extent, small bowel FDG uptake and may mask a lesion.
- *Heart:* Uptake is often variable, depending on insulin/glucose levels and the patient's fasting state. The heart prefers fatty acids but will use glucose in the postprandial state.
- *Muscles:* FDG uptake is normally low. However, elevated insulin levels or recent exercise can cause increased muscle uptake.
- *Brown fat:* **Brown fat** is metabolically active adipose tissue that can be found in the supraclavicular, axillary, mediastinal and paravertebral regions in adults that can be mildly to moderately FDG avid, especially if the patient is cold or after food intake.

Oncologic indications of F-18 FDG PET/CT

Lung cancer



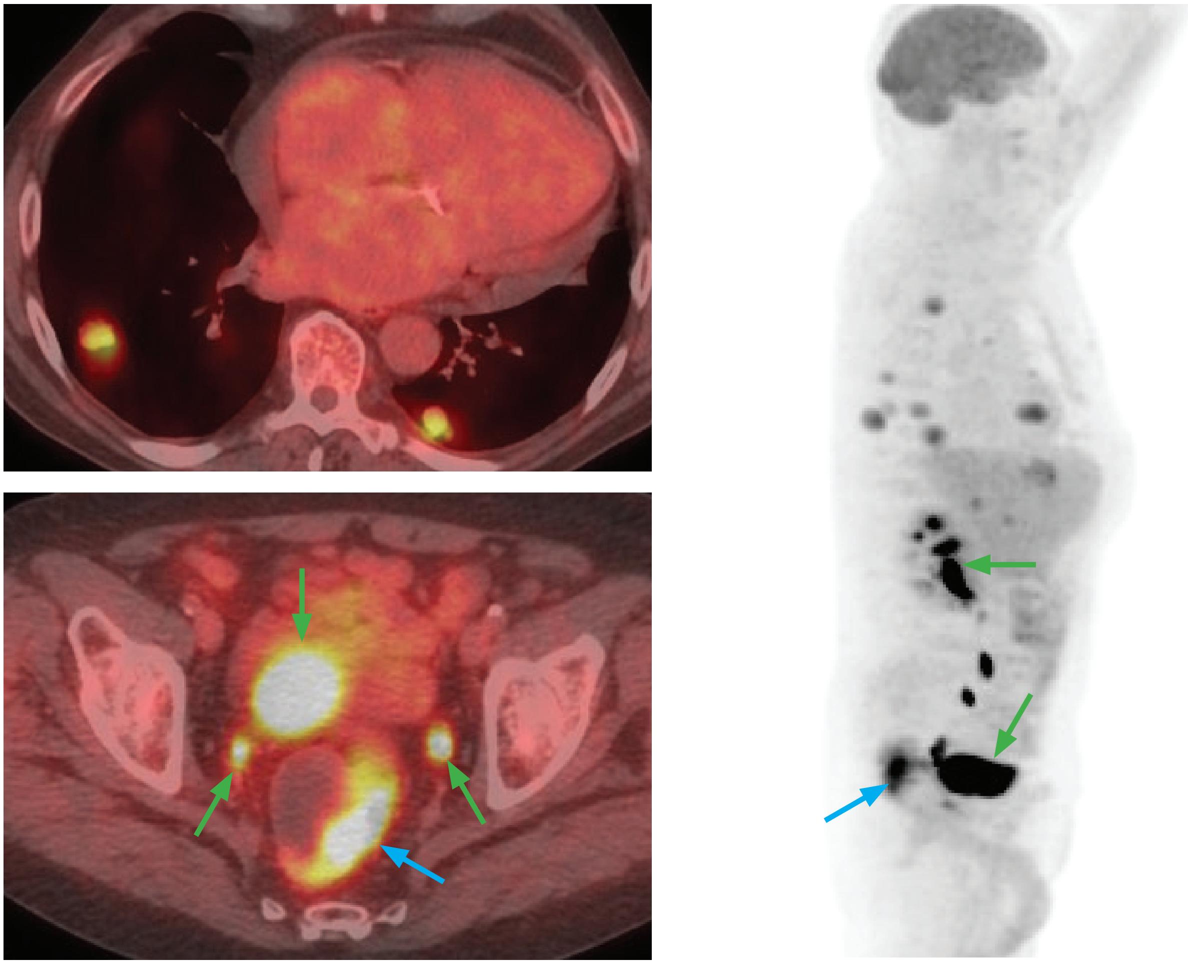
Lung cancer initial staging: Fused axial PET/CT (left image) and coronal PET MIP (right image) show an intensely FDG-avid right upper lobe mass (blue arrow) representing the patient's primary bronchogenic carcinoma. There were no enlarged or FDG-avid hilar or mediastinal lymph nodes, and no FDG-avid distant metastases. Also note normal biodistribution of FDG in the other organs.

- PET/CT plays a role both in the initial staging of patients with lung cancer, and in evaluating response to treatment.
- For initial staging, PET/CT is most useful in evaluating local tumor extension and searching for distant metastases. Approximately 10% of patients with a negative metastatic workup by CT will have PET evidence of metastasis.
- PET/CT also plays a role in the initial detection of mediastinal and hilar lymph nodes. Although PET is very sensitive for detecting malignant lymph nodes, it is not specific. Mediastinoscopy is the gold standard for lymph node staging. Given the lack of PET specificity, PET-positive mediastinal nodes must be followed by mediastinoscopy before potentially curative surgery is denied based solely on the PET.

Solitary pulmonary nodule (SPN)

- Evaluation of a suspicious solitary pulmonary nodule (SPN) is an indication for PET/CT.
- Depending on the equipment used, 5–8 mm is typically the smallest size for a nodule that can be reliably evaluated by PET.
- The majority of malignant SPNs are FDG-avid. However, low-grade tumors such as lung adenocarcinoma or carcinoid may not be metabolically active and may be falsely negative on PET.
- Conversely, the majority of benign SPNs are not FDG-avid. Active granulomatous disease (including tuberculosis) may take up FDG, thus falsely positive on PET.
- It is never possible to definitively diagnose a nodule as benign or malignant based on SUV value alone.
- In general, if a nodule is not FDG-avid, short-term follow-up is reasonable. If the nodule is FDG-avid then biopsy or resection is preferred.

Colon cancer



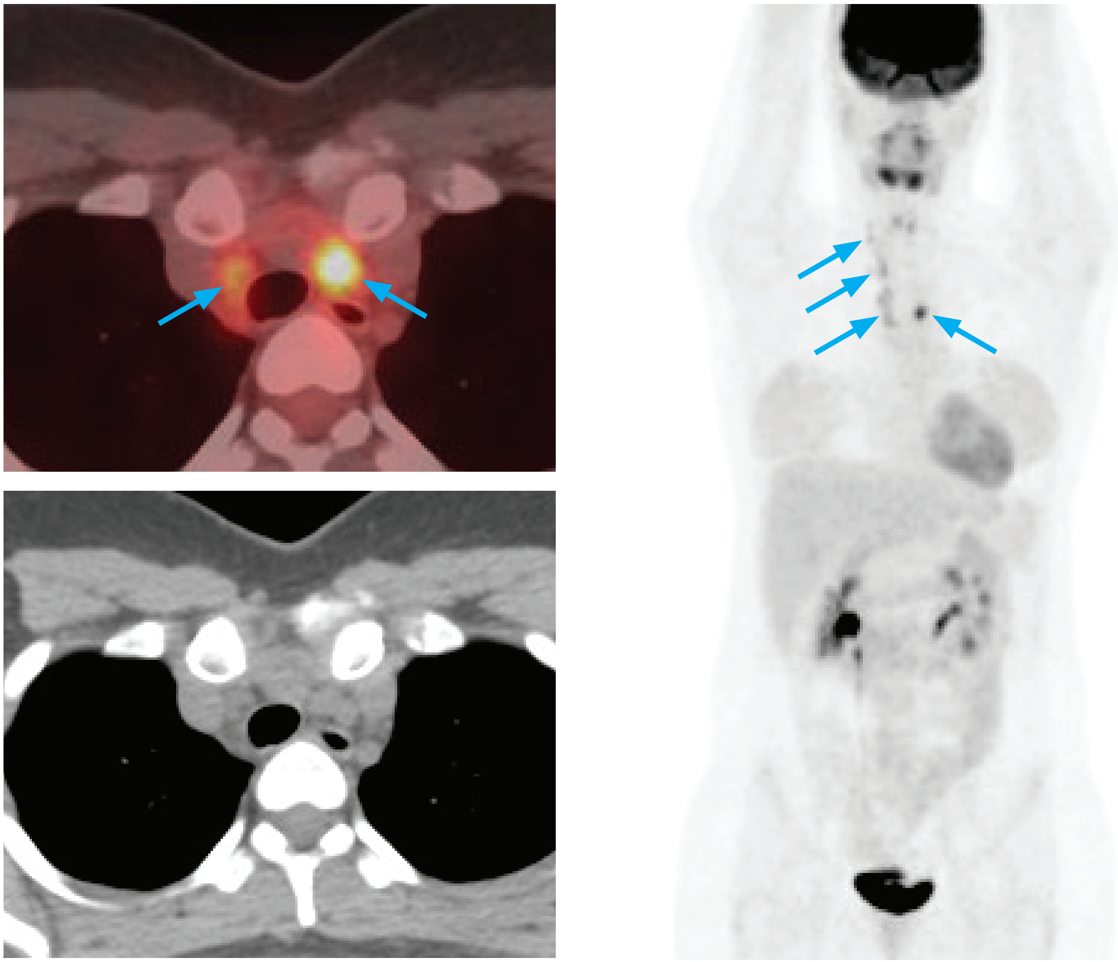
Metastatic rectosigmoid cancer: Fused axial PET/CT through the lungs (top left image), through the pelvis (bottom left image), and a sagittal PET MIP (right image) show the intensely FDG-avid primary rectosigmoid malignancy (blue arrows), with numerous pulmonary metastases. Note the physiologic intense activity in the renal collecting system, bladder and ureters (green arrows).

- PET/CT has a limited role in determining local extent of colon cancer due to poor spatial resolution and physiologic bowel uptake, but it does play a primary role in evaluating metastatic disease in colorectal malignancies. In particular, since an isolated hepatic metastasis can be resected or ablated, evaluation for extrahepatic metastases is a common indication for PET/CT.
- After initial treatment, follow-up PET/CT is usually delayed approximately two months due to increased FDG uptake in the peritreatment period related to inflammation that can occur.

Head and neck cancer

- PET/CT is often used in the initial staging of head and neck squamous cell carcinoma, especially for evaluation of regional nodal metastases.
- Specificity for evaluating recurrent disease after chemoradiation is limited due to altered anatomy and inflammation from treatment. Usually post-treatment scans should be delayed 4 months after treatment.

Thyroid cancer



Iodine-resistant thyroid cancer: Fused axial PET/CT (top left image), non-contrast CT (bottom left image), and coronal PET MIP show several moderately FDG-avid right cervical and paratracheal lymph nodes and a single intensely FDG-avid left paratracheal node (arrows). There is symmetric physiologic salivary gland uptake seen on the coronal MIP. I-131 scan (not shown) was negative.

- Undifferentiated or medullary thyroid cancers may not take up radioiodine but may be FDG avid. PET/CT is used in the clinical setting of a rising thyroglobulin level with negative whole-body radioiodine scans.

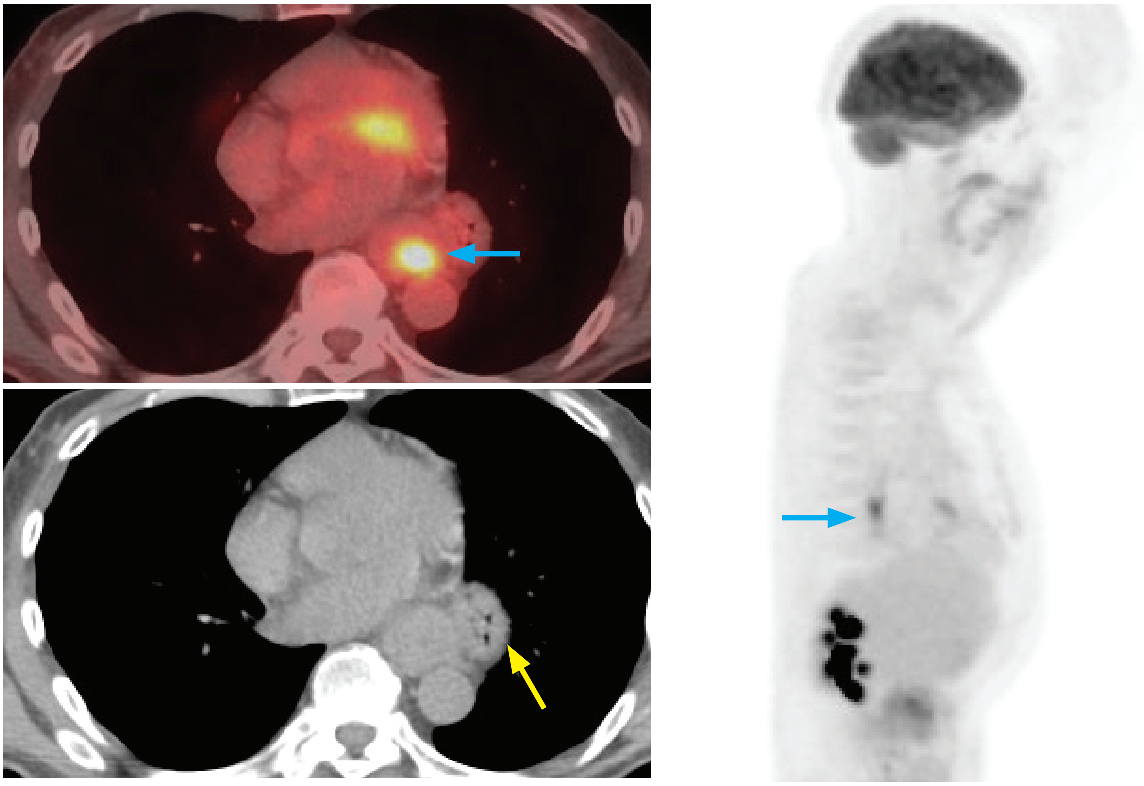
Lymphoma

- PET/CT plays a key role in the staging and restaging of patients with lymphoma.
- Most histological types of lymphoma, including Hodgkin and non-Hodgkin lymphoma, are FDG-avid. Low-grade lymphomas, such as small lymphocytic and mantle cell lymphoma, however, tend to be less FDG-avid.
- Increased marrow uptake in lymphoma patients can be difficult to interpret. Diffuse marrow uptake may be due to granulocyte colony-stimulating factor (G-CSF) stimulation, rebound effect from chemotherapy, or malignant marrow infiltration. Focal increased uptake, however, is more likely to represent lymphoma.

Breast cancer

- Although used in the staging and response to therapy of recurrent or stage IV breast cancer, standard PET/CT is not used to evaluate small nodal metastases.
- PET/CT is not routinely used for patients with stage I–III breast cancer.

Esophageal cancer



Non-metastatic esophageal cancer: Fused axial PET/CT (top left image) and sagittal PET MIP (right image) show a single focus of intense FDG uptake within the distal esophagus (blue arrows). Marked esophageal thickening is also apparent on the non-contrast CT (bottom left image). There is no evidence of metastatic disease. The focus of FDG uptake within the heart on the fused image is physiologic myocardial uptake. Note the presence of a small paraesophageal hernia (yellow arrow).

- The primary role of PET/CT in the evaluation of esophageal cancer is to identify patients with stage IV disease who are not surgical candidates. PET/CT is limited for primary tumor staging of esophageal cancer, which is better assessed with endoscopic ultrasound.
- After initial neoadjuvant treatment, a decrease in FDG avidity by at least 30% suggests a more favorable prognosis. In contrast, those patients who do not show a decrease in FDG uptake can potentially be spared ineffective chemotherapy regimens.

Cancers where F-18 FDG PET/CT plays a limited role

Hepatocellular carcinoma (HCC)

- Only 50% of HCCs can be imaged with FDG-PET due to relatively high levels of phosphatase compared to other cancers which dephosphorylates FDG and allows it to diffuse out of cells.

Renal cell carcinoma (RCC) and bladder cancer

- Only 50% of RCCs are FDG-avid, though PET may play a role in detecting metastatic disease.
- Detection of ureteral or bladder lesions is extremely limited due to surrounding high urinary FDG activity.

Prostate cancer

- Most prostate cancers have low FDG uptake.
- FDG-PET is reserved for assessing treatment response of bony metastases in aggressive castration-resistant prostate cancer and as a prognostic indicator.

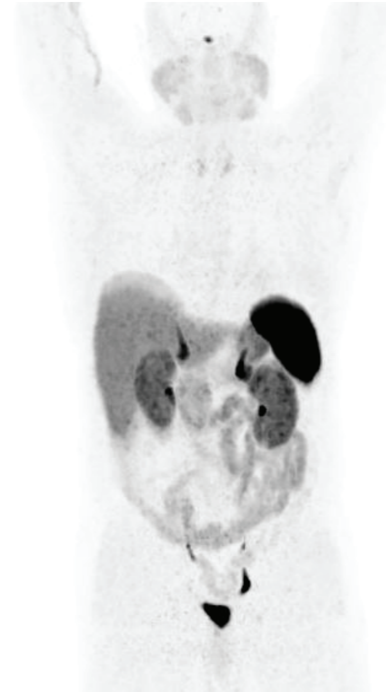
Other tumors

- Well-differentiated or mucinous cell type tumors, lung adenocarcinoma, and carcinoid tumors tend to have low FDG uptake or are non-FDG-avid.

Other types of PET/CT

Gallium-68 DOTATATE (GaTate)

- Gallium-68 is a generator-produced positron emitter with a half-life of 68 minutes.
- Ga-68 DOTATATE is a somatostatin analog that is used in the imaging of tumors with high somatostatin receptor (SSTR) expression, including neuroendocrine tumors, pheochromocytoma, paraganglioma, neuroblastoma, and meningioma.
- DOTATATE PET/CT is the new first-line diagnostic imaging modality for neuroendocrine tumors, replacing the older Octreoscan and MIBG scintigraphy (discussed later in this chapter).
- DOTATATE and FDG PET/CT are complementary exams, helping to identify both well-differentiated (high DOTATATE uptake) and poorly differentiated (high FDG uptake) tumors.



Coronal PET MIP shows physiologic distribution of Ga-68 DOTATATE. Note intense uptake in the spleen, followed by the adrenals, kidneys/bladder, pituitary, salivary glands, thyroid, liver, and bowel.

PET agents for prostate cancer imaging

- Because of the limitation of FDG PET/CT in prostate cancer imaging, a number of alternative PET radiotracers have been developed in the past decade.
- **Carbon-11/F-18 choline** are biomarkers of cell membrane metabolism and are FDA-approved for imaging biochemically recurrent prostate cancer with high PSA levels, but are not in widespread use.
- **PSMA ligands** such as ProstaScint and Ga-68 PSMA target the prostate membrane specific antigen (PSMA) which is highly expressed in prostate cancer. PSMA PET/CT is highly sensitive for biochemically recurrent prostate cancer with low PSA levels, but is not yet FDA-approved.
- **F-18 fluciclovine (Axumin)** is an amino acid analog which targets amino acid transporters that are overexpressed in prostate cancer. Imaging is performed only minutes after injection of the tracer, which limits any significant urinary excretion. It is FDA-approved for detection of biochemically recurrent prostate cancer.
- **F-18 sodium fluoride** (F-18 NaF) is an analog of the hydroxyl group in hydroxyapatite bone crystals and a biomarker of osteoblastic activity. It is highly sensitive for the detection of prostate cancer bone metastases, but is not routinely used due to low specificity, high cost, and limited availability.

Coronal PET MIP shows normal distribution of F-18 fluciclovine. Note intense uptake in liver and pancreas, and variable uptake in muscles. Foci of uptake at the right arm represent the injection site.

Cerebrovascular

Radiotracers

Tc-99m DTPA / Tc-99m pertechnetate

- Technetium-99m has a half-life of 6 hours and is used in the majority of nuclear imaging.
- Tc-99m DTPA and pertechnetate are transient perfusion agents. There is no uptake within the brain parenchyma as they do not cross intact blood-brain barrier. Activity is seen in the scalp, calvarium, subarachnoid spaces, major cerebral arteries and venous sinuses.
- Transient perfusion agents are reserved for planar imaging only and are now uncommonly used.

Tc-99m HMPAO / Tc-99m ECD

- Both HMPAO and ECD are lipophilic perfusion agents that cross the blood-brain barrier. To be retained in the cell, ECD is enzymatically modified. In contrast, HMPAO simply needs to be protonated to be trapped. Thus, ECD is only taken up by living cells while HMPAO uptake is a marker of perfusion. In the evaluation of subacute infarct, the phenomenon of luxury perfusion can cause HMPAO uptake to increase (due to increased perfusion), while ECD will show the true defect representing the infarct core.
- HMPAO and ECD show activity in the brain parenchyma, mainly gray matter.
- Of these two agents, ECD is generally preferred for brain imaging, because it has more rapid blood pool clearance, better shelf life, more accurate characterization of perfusion, and is only taken up by living cells.
- Both tracers are used for SPECT imaging.

Iodine-123 ioflupane (I-123 FP-CIT or DaTscan)

- Iodine-123 has a half-life of 13 hours. I-123 ioflupane is a cocaine analog that binds to the presynaptic dopamine transporter (DaT). It is used in SPECT imaging for the differentiation of essential tremor from parkinsonism.

Thallium-201

- Thallium-201 has a half-life of 73 hours and is used for SPECT imaging in the differentiation of tumor recurrence versus radiation necrosis. Thallium accumulates in viable tumor but not normal brain tissue.

PET agents

- F-18 FDG is the most common FDA-approved PET agent used for brain imaging.
- F-18 florbetapir, F-18 florbetaben and F-18 flutemetamol are FDA-approved PET agents for imaging of beta-amyloid.
- In 2020, F-18 flortaucipir was approved by FDA for tau imaging.
- Other PET radiotracers are emerging that target specific neuroreceptors (such as F-18-DOPA targeting the dopamine transporter).

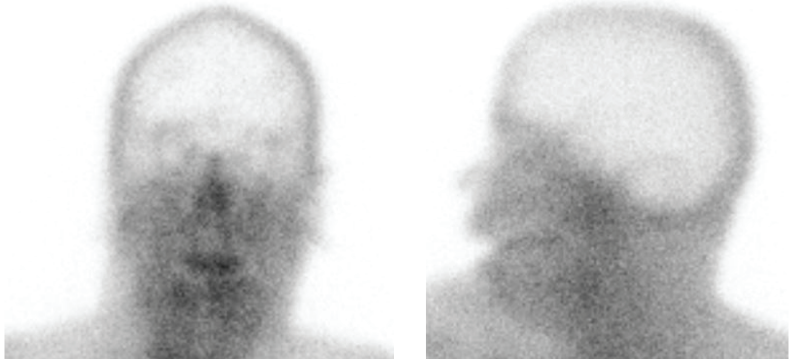
Radionuclide imaging of the brain

Evaluation of cerebral perfusion

- Cerebral perfusion reserve can be assessed with an **acetazolamide (Diamox) challenge**. Normally, cerebral blood flow increases after administration of 1 g Diamox, a carbonic anhydrase inhibitor that causes increased cerebral CO_2 concentration and thereby cerebrovascular vasodilation.

- Areas of the brain that already have maximized their autoregulatory mechanisms will not show an increase in perfusion after Diamox administration, and therefore will have relatively reduced activity compared to the rest of the brain. These areas are at risk for infarction.

Brain death

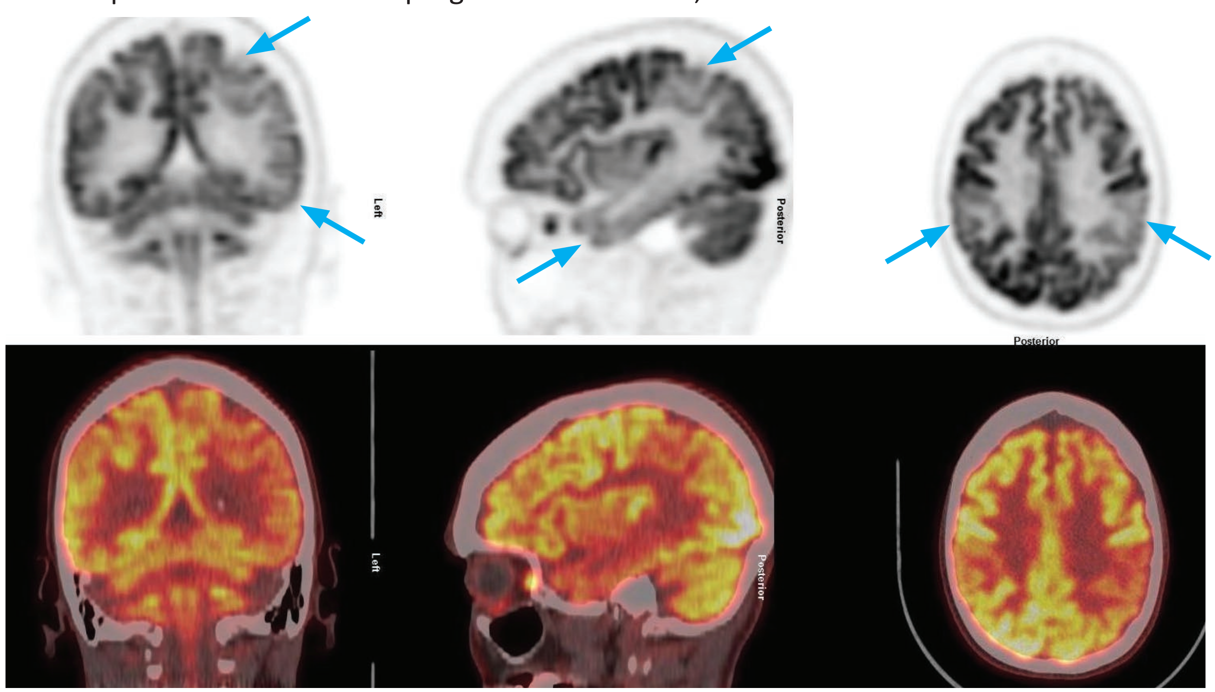


Frontal and lateral planar images following injection of Tc-99m HMPAO show absent intracranial uptake (*empty light bulb sign*) and increased perfusion in the face (particularly nasal region) and scalp. Absent cerebral perfusion is consistent with brain death in the proper clinical setting.

- In brain death, intravenously injected radiotracer is unable to enter the cranial cavity due to increased intracranial pressure.
- SPECT imaging can be performed, either with HMPAO or ECD-labeled Tc-99m.
- Imaging shows absence of tracer perfusing the brain. The *hot-nose sign* represents increased collateral flow seen in brain death, but is not specific.

Dementia imaging

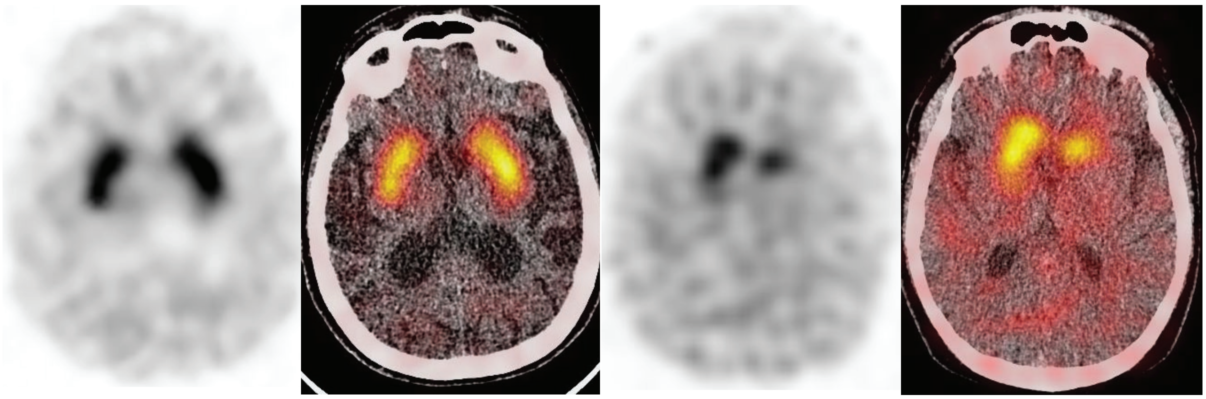
- Perfusion imaging with SPECT and glucose metabolism with PET are helpful in the early diagnosis of dementia based on parallel patterns.
- **Alzheimer disease** typically shows asymmetrically to symmetrically reduced perfusion (on SPECT) or hypometabolism (on FDG-PET) in the posterior cingulate gyrus and posterotemporal and parietal lobes, with sparing of the sensorimotor cortex, occipital lobes and deep nuclei. With further progression of disease, frontal lobes can also be involved.



F-18 FDG PET and PET/CT overlay images of a patient with memory difficulties show hypometabolism predominantly in the parietal and temporal cortex (arrows), greater on the left. There is sparing of the sensorimotor cortex, occipital cortex, and deep nuclei (not shown). This pattern can be seen with Alzheimer disease in the appropriate clinical context. *Case courtesy Matthew Robertson, MD.*

Dementia imaging (continued)

- **Lewy body dementia** appears similar to Alzheimer but also involves the occipital lobes.
- **Multi-infarct dementia** shows multiple asymmetric foci of decreased metabolism/perfusion.
- **Pick disease** (frontotemporal dementia) is characterized by decreased uptake in the frontal lobes and anterior portion of the temporal lobes.
- **Huntington disease** shows decreased uptake in the basal ganglia.
- **Parkinson disease** is characterized by decreased dopamine transporter density in the striatum on SPECT and PET, demonstrated with use of agents such as I-123 ioflupane (DaTscan) or F-18-DOPA.



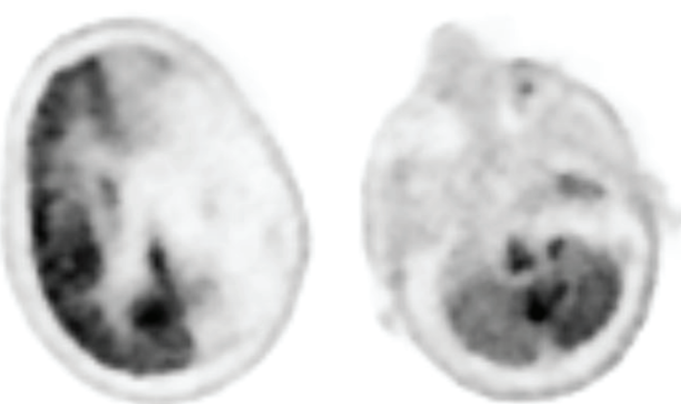
Normal versus abnormal I-123 ioflupane uptake on SPECT and SPECT/CT overlay: The left two images show normal symmetric comma-shaped activity in the caudate nucleus and putamen. The right two images show asymmetric and reduced activity in both striata, greater on the left, resulting in loss of the comma shape.

Seizure imaging

- In general, blood perfusion and metabolism of seizure foci are increased during seizure (ictal imaging), and decreased between seizures (interictal imaging).
- Ictal perfusion SPECT with Tc-99m HMPAO or ECD is the most sensitive, but technically challenging to perform because the radiotracer must be injected during the seizure or within 30 seconds after the end of the seizure.
- Interictal PET with F-18 FDG is very sensitive in localizing partial complex seizures, which usually originate in the temporal lobe. Ictal PET is not possible due to slow FDG uptake.
- Extratemporal seizure focus is more difficult to localize. Ictal SPECT is the most sensitive.

Crossed cerebellar diaschisis

- Crossed cerebellar diaschisis is a commonly encountered phenomenon in the presence of a supratentorial lesion (seen in tumors, stroke, and trauma), where the cerebellar hemisphere contralateral to the lesion shows decreased radiotracer uptake. This phenomenon is thought to be due to interruption of corticopontine-cerebellar pathways.

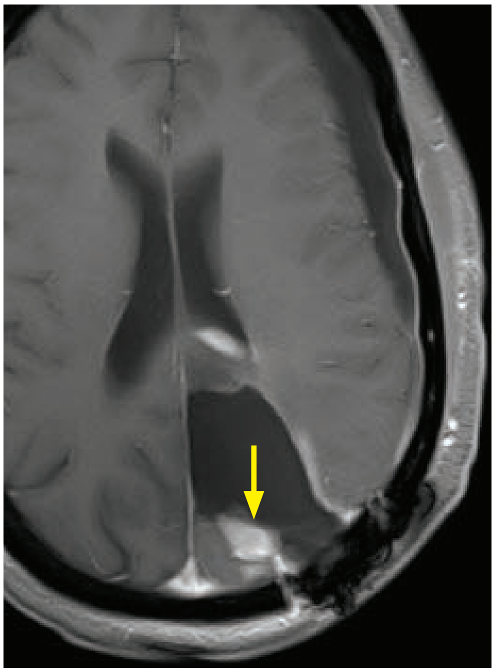


Axial F-18 FDG PET (right image) shows near absent glucose metabolism in the left cerebral hemisphere due to known encephalomalacia. Relatively decreased FDG uptake in the *right* cerebellum (left image) is consistent with crossed cerebellar diaschisis.

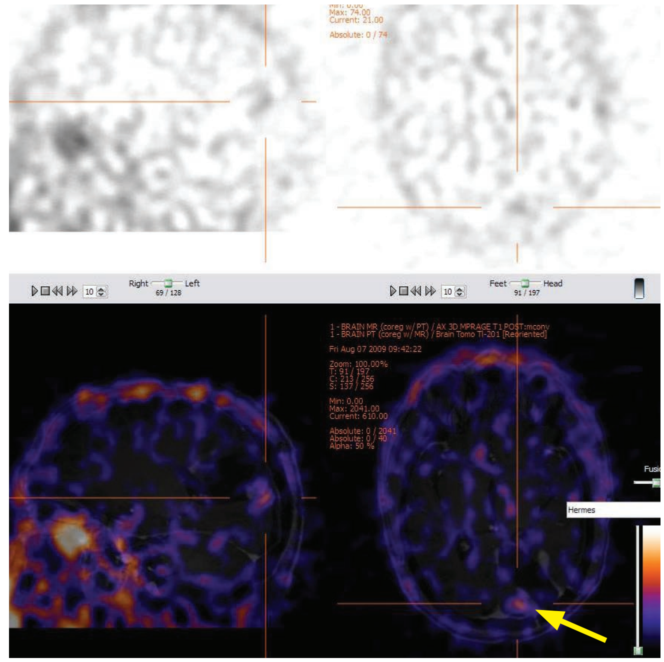
Brain tumor recurrence versus radiation necrosis

- Several nuclear imaging protocols have been described in the differentiation of tumor recurrence versus radiation necrosis. Regardless of the radiotracer involved, fusion to MRI is usually performed.
- Thallium-201 generally accumulates in malignant gliomas and not in post-treatment granulation tissue (i.e., thallium is not taken up by radiation necrosis). Thallium-201 uptake requires a living cell and blood-brain barrier disruption. The degree of uptake can be graded in comparison to the scalp activity:

  Uptake <scalp is low; uptake between scalp and 2x scalp is moderate; and uptake >2x scalp is high.



Axial T1-weighted post-contrast MRI shows a left parietal resection cavity with an enhancing nodule (arrow) along the posteromedial aspect of the resection cavity.



SPECT thallium-201 (top image, with 4.4 mCi administered) and fused SPECT-MRI (bottom image) show thallium uptake in the posteromedial aspect of the resection cavity correlating with the enhancing nodule seen on MRI (arrow). This pattern is suggestive of tumor recurrence.

- Dual-phase F-18 FDG PET employs early and delayed imaging to evaluate a region of suspected tumor recurrence versus radiation necrosis. PET scanning is performed at 1 hour and 4 hours after the injection of the radiotracer. Dual phase PET has been shown to have increased accuracy for the assessment of recurrence versus post-treatment changes in metastatic disease compared to single-phase PET.

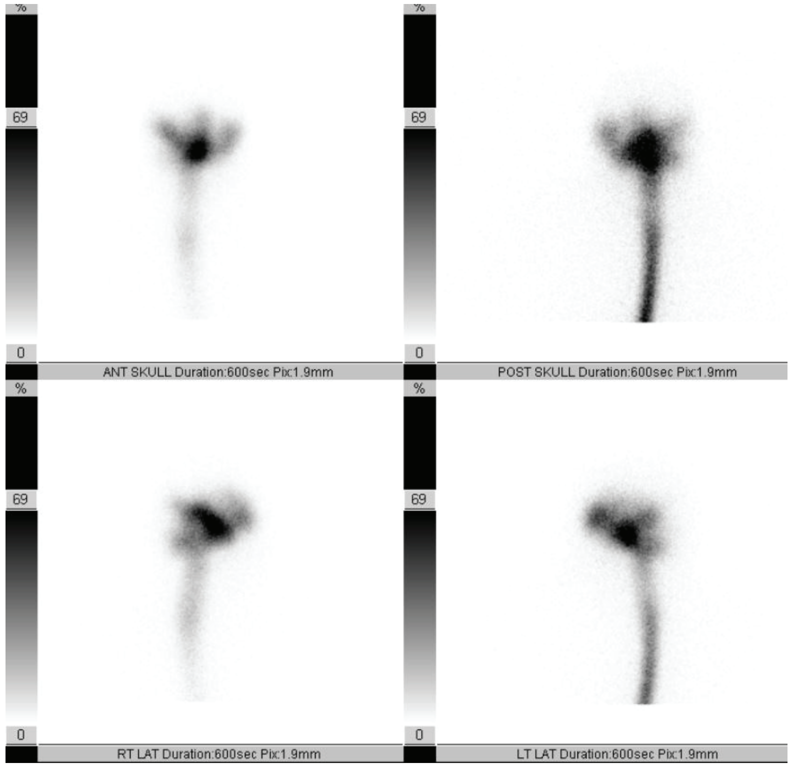
Cerebrospinal fluid imaging

Radiotracer: Indium-111 DTPA

- Indium-111 has a half-life of 67 hours.
- In-111 DTPA is ideal for cerebrospinal fluid imaging because it is non-lipid soluble, not metabolized, and has a long half-life (CSF imaging may span several days). It is administered intrathecally by lumbar puncture.
- Normally, radiotracer should reach the basal cisterns by 1 hour, the sylvian and anterior interhemispheric cisterns by 2–6 hours, cerebral convexities by 12 hours, and the arachnoid villi in the sagittal sinus by 24 hours.
- Presence of radiotracer activity in the lateral ventricles is abnormal. Failure of radiotracer ascent over the cerebral convexities at 24 hours is also abnormal.

Radiotracer: Tc-99m DTPA

- Due to its short half-life (6 hours), Tc-99m DTPA is used for CSF studies that do not require prolonged imaging, including the evaluation of CSF leak and shunt patency.



Normal cisternogram:

Planar images at different projections were obtained at 4 hours following intrathecal injection of 0.54 mCi In-111 DTPA. There is normal *trident appearance* of tracer activity in the sylvian and anterior interhemispheric cisterns. Activity is also present in the basilar cisterns and spinal canal.

Normal pressure hydrocephalus

- Radionuclide cisternography can be used to confirm suspected communicating hydrocephalus in patients with the classic clinical triad of normal pressure hydrocephalus (ataxia, incontinence, dementia), who would benefit from CSF shunting.
- Communicating hydrocephalus is characterized by early entry of radioactivity into the lateral ventricles within 6 hours, persistent intraventricular activity, and lack of activity over the cerebral convexities after 24–48 hours.

CSF leak

- Common locations for CSF leak are the ear, paranasal sinuses, and nose.
- CSF leak can be identified by asymmetric activity in the region of the ears on frontal imaging, or activity in the nose on lateral imaging. Subtle leaks can only be detected by placing cotton pledgets in the area of concern, removing after several hours and counting the pledgets' activity relative to plasma activity.
- For spinal CSF leak, radioisotope cisternography shows diffusion of the tracer into the paraspinal extra-arachnoidal space. Correlation with MR or CT myelography may show triangular CSF space expansion at nerve root sleeves. A common pitfall is perineural cysts which are discrete round lesions.

Evaluation of CSF shunt patency

- Tc-99m DTPA or less commonly In-111 DTPA are injected into the shunt reservoir.
- If the distal limb of the shunt is patent, radiotracer activity should rapidly pass through the shunt catheter and into the peritoneal cavity or right atrium.
- Manual occlusion of the distal limb should normally show reflux of radiotracer into the ventricular system. This is not seen in the presence of proximal limb occlusion.

Thyroid

Radiotracers

Iodine-131

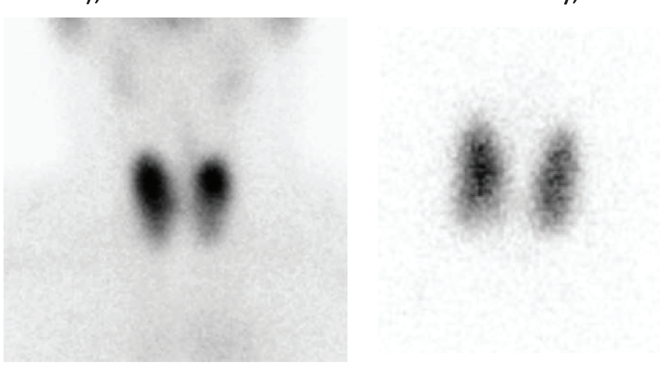
- I-131 emits both beta particles and 364 keV gamma photons (only the gamma photons are used for imaging). The half-life is 8 days. I-131 is produced by fission in a nuclear reactor. It is administered orally.
- I-131 is ideal for therapy due to its high radiation dose to the thyroid and relatively low whole body dose. Indications include treatment of thyroid cancer status post thyroidectomy, and hyperthyroidism from Graves disease or multinodular goiter.

Iodine-123

- I-123 decays by electron capture and emits 159 keV gamma photons. The half-life is 13 hours. I-123 is expensive because it is produced by cyclotron. It is administered orally.
- I-123 is the preferred radioisotope for thyroid imaging, as it can image in high detail and obtain thyroid uptake values.

Tc-99m pertechnetate

- Tc-99m emits 140 keV gamma photons, has a half-life of 6 hours, and is generator-produced.
- Unlike iodine, pertechnetate is trapped but not organified by the thyroid. After initial uptake it is released into the blood pool. Thyroid uptake is not routinely quantified with pertechnetate due to its rapid washout.
- Pertechnetate is an excellent alternative to I-123 for thyroid imaging because of its ready availability from generators, and its low patient dose which allows administration of higher doses and thereby permits quicker imaging with less motion artifact. It is administered intravenously.
- Because pertechnetate does not specifically localize to the thyroid, high background counts are typical. Only 1–5% of administered activity is taken up by the thyroid.
- In contrast to I-123, the salivary glands are well seen with pertechnetate.
- Tc-99m pertechnetate is preferred over I-123 when the patient has received recent intravenous iodinated contrast (iodine in contrast blocks thyroid uptake of additional iodine), when IV administration is necessary, or when a quick study is required.



Tc-99m pertechnetate (left image) versus I-123 (right image) thyroid scans. Note, pertechnetate has higher background counts and physiologic uptake in salivary glands.

Patient preparation

- Patients undergoing I-123 or I-131 imaging/therapy must have non-suppressed TSH, which can be achieved by stopping exogenous thyroid hormone for 4 weeks, or by two intramuscular injections of recombinant TSH (rTSH).
- Patients who received intravenous iodinated contrast should wait one month before radioiodine imaging.

Pregnancy and breastfeeding

- All thyrotropic agents cross the placenta and I-131 is contraindicated in pregnancy. Fetal iodine is taken up beginning at 12 weeks of gestation.
- A breastfeeding mother who requires an I-131 ablative dose must stop breastfeeding permanently for the current child.
- For I-123, breastfeeding can be resumed two to three days after administration.
- For Tc-99m, breastfeeding can be resumed 12–24 hours after administration.

Diagnostic Indications (I-123 or Tc-99m pertechnetate)

Ectopic thyroid

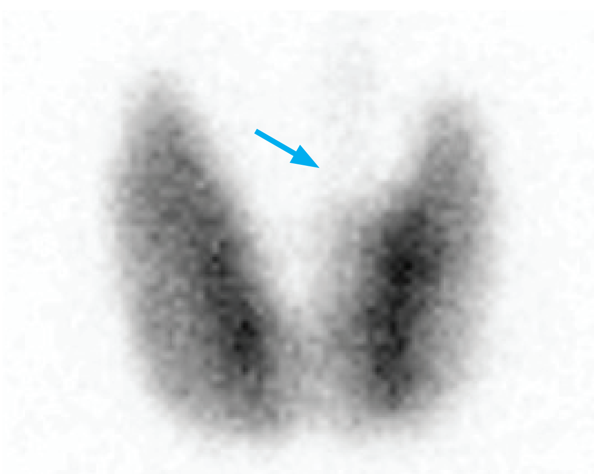
- Either I-123 or Tc-99m can be used to localize suspected ectopic thyroid tissue.
- Lingual thyroid is ectopic thyroid tissue at the base of the tongue.
- Functional thyroid tissue may rarely be seen in an ovarian teratoma (struma ovarii).
- Retrosternal thyroid tissue is most often due to a substernal goiter.

Thyroid nodule

- Thyroid nodules are typically only imaged if the cytology is indeterminate.
- Hyperfunctioning nodules are almost always benign adenomas.
- Cold nodules have approximately 20% risk of malignancy, although the most common cold nodule (~70–75%) is a benign colloid cyst.
- A warm nodule usually represents a cold nodule with overlapping thyroid tissue.

  A warm nodule requires further investigation such as biopsy if oblique views are indeterminate.
- A discordant thyroid nodule is "hot" on Tc-99m and "cold" on I-123 as it has maintained the ability to trap pertechnetate but is unable to organify iodine. Biopsy is usually recommended as a discordant nodule may be malignant.

Graves disease



Graves disease on I-123 scan with 0.3 mCi I-123 NaI administered.

There is diffuse uptake throughout the thyroid gland. A faint pyramidal lobe is present (arrow).

24-hour uptake is elevated at 56%.

- Graves disease is an autoimmune disorder characterized by hyperthyroidism, thyromegaly, homogeneously increased thyroid activity, and often a prominent pyramidal lobe.
- Both 6-hour and 24-hour **iodine uptake** are elevated.

  Normal 6-hour uptake is 6–18% and normal 24–hour uptake is 10–30%.
- Although usually an I-123 and a Tc-99m scan can be differentiated by the presence of salivary uptake with Tc-99m, in Graves disease this distinction is often not possible. In Graves disease, thyroid uptake can be so strong that the salivary glands are often not seen, causing a similar appearance with either radiotracer.

Graves disease (continued)

- Definitive treatment of Graves disease is I-131 radiotherapy or (less commonly) surgery. Antithyroid drugs (e.g., methimazole or propylthiouracil) are another option and may achieve a remission after one to two years of use.

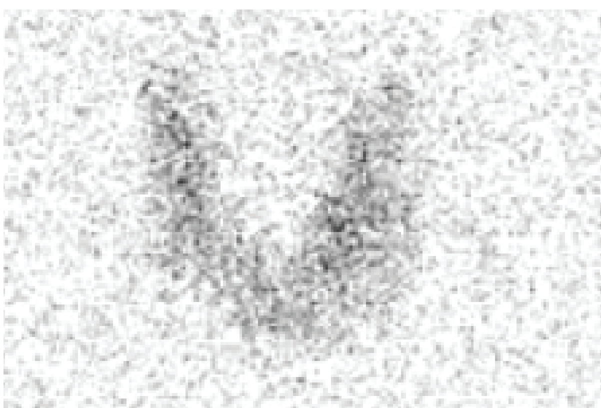
Toxic multinodular goiter (Plummer disease)

- Patients with toxic multinodular goiter are often middle-aged to elderly women and present with symptoms of hyperthyroidism.
- Thyroid scan shows an irregular, nodular thyroid contour and heterogeneous uptake corresponding to single or multiple hot nodules with suppression of the remaining normal functioning thyroid. Iodine uptake is elevated.
- Treatment options include medical therapy with antithyroid drugs for thyrotoxicosis, I-131 ablation, or thyroidectomy.
- Malignancy is relatively uncommon in a multinodular goiter. While a dominant cold nodule should undergo further investigation, smaller cold nodules are unlikely to be malignant.

Hashimoto thyroiditis

- Hashimoto thyroiditis is the most common inflammatory disease of the thyroid.
- Like Graves disease, Hashimoto thyroiditis also clinically presents with thyromegaly. In Hashimoto thyroiditis, however, thyroid hormone levels are variable depending on the disease stage. Most patients with Hashimoto thyroiditis are hypothyroid.
- Appearance on thyroid scan is variable, ranging from diffusely increased activity that resembles Graves disease to patchy uptake similar to a multinodular goiter. The patchiness is thought to be due to cold areas from infiltration by lymphocytes and lymphoid follicles.

Subacute thyroiditis



Subacute thyroiditis on I-123 scan with 0.22 mCi administered.

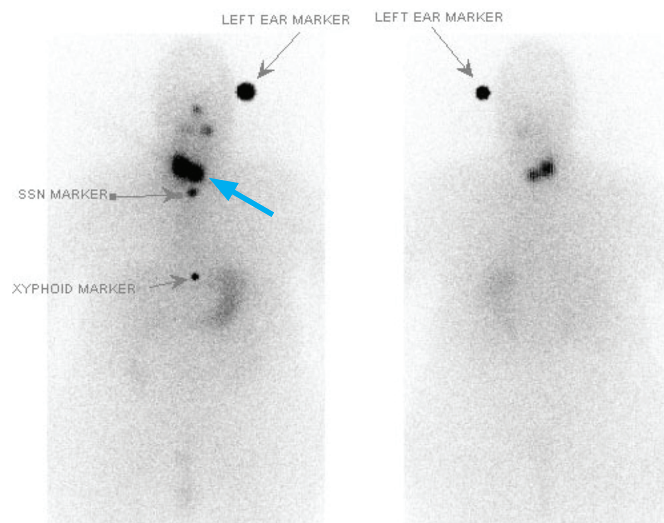
There is diffusely low thyroid uptake with very low background to thyroid uptake ratio. 24-hour uptake was only 4.9%.

- The classical clinical presentation of subacute thyroiditis is a painful swollen gland, although many patients present with silent hyperthyroidism.
- Imaging shows decreased radiotracer uptake and a low 24-hour uptake.
- Subacute thyroiditis is typically a self-limited condition. Treatment is directed towards symptom control with nonsteroidal anti-inflammatory drugs or steroids in severe cases.

Therapeutic indications (I–131)

Thyroid carcinoma, post-thyroidectomy

- Approximately one to two months after thyroidectomy, I-131 is administered to treat and simultaneously image residual and potential metastatic disease.
- Following thyroidectomy, thyroid replacement therapy is withheld to allow endogenous TSH to increase. This increases uptake of therapeutic I-131 by any residual or metastatic thyroid tissue. The goal TSH is 30–50 mIU/mL.
- The dosing of I-131 is dependent on the oncologic risk:
  - Low-risk patient (tumor <1.5 cm, no invasion of thyroid capsule): ≤30 mCi I-131 administered.
  - High-risk patient: 100–200 mCi I-131 administered.



I-131 ablative therapy: 99 mCi I-131 administered orally 4 days prior to scanning. The gray arrows point to markers.

Anterior and posterior planar imaging shows a large focus of residual uptake in the thyroid bed (arrow) and mild symmetric salivary gland uptake. There is physiologic uptake in the stomach, bladder, and bowel.

- A new approach is a standard dose of 30 mCi for treatment of all T1, T2, and N1 cancers.
- Functioning lung or skeletal metastases require high doses of I-131, usually >200 mCi.
- To avoid or minimize the complication of irreversible pulmonary fibrosis, care is taken to limit the whole-body retention to 80 mCi at 48 hours after administration. To avoid bone marrow suppression, the absorbed dose to the blood is limited to 2 Gy.

Thyroid carcinoma, post radioiodine therapy

- After ablation with I-131, patients with thyroid carcinoma are monitored by following thyroglobulin levels. If thyroglobulin levels rise, an I-123 scan is performed to evaluate for disease recurrence or metastasis. If the I-123 scan is positive, repeat I-131 radioiodine is administered for ablation. Note that the presence of anti-thyroglobulin antibodies precludes the ability to monitor the thyroglobulin levels.

Treatment of Graves disease

- I-131 is administered in a single oral dose to treat Graves disease. Contraindications to I-131 include pregnancy, lactation, and inability to comply with radiation safety guidelines.
- The dosing of I-131 for treatment of Graves varies by institution. Many endocrinologists advocate a calculated dose based on the estimated thyroid weight and 24-hour uptake, while another study has shown one of three fixed doses (up to 15 mCi) to be equally effective.
- Adequate dosing of I-131 can treat greater than 90% of patients with Graves disease.

Treatment of multinodular goiter

- I-131 can be used to reduce goiter size, especially in patients who choose not to undergo surgery. Compared to Graves disease, toxic multinodular goiter is more resistant to iodine radiotherapy, therefore requires a higher I-131 dose (20–30 mCi) and often multiple treatments.

Treatment of toxic/autonomous nodule

- Solitary toxic thyroid nodules can be successfully treated with 20–25 mCi of I-131.

Parathyroid

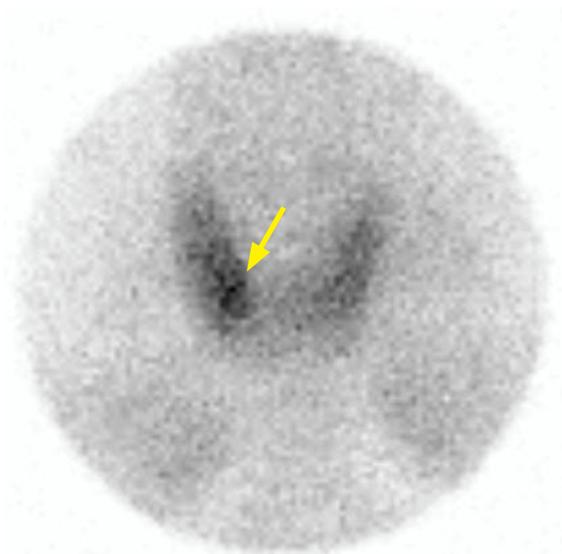
Radiotracer

Tc-99m sestamibi

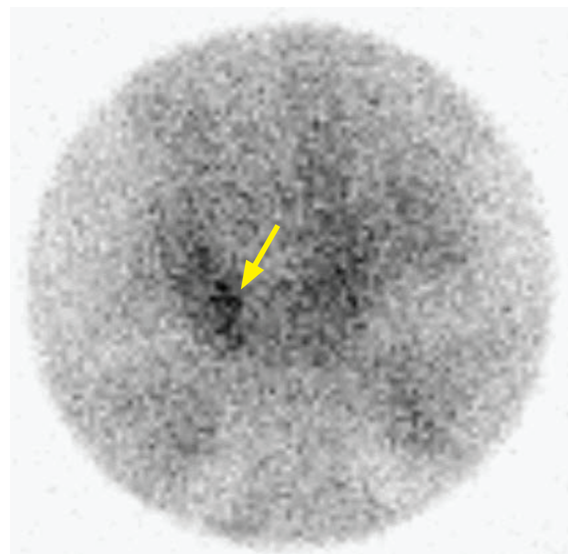
- Tc-99m sestamibi localizes to the mitochondria in parathyroid adenomas, hyperplastic parathyroids, normal thyroid tissue, and thyroid adenomas; however, thyroid activity decreases over time. It is *not* taken up by normal functioning parathyroid tissue.
- The same agent is used for nuclear cardiology and can also be taken up by malignancy, particularly breast and lung cancers.

Indications

Parathyroid adenoma



Early anterior pinhole image of the thyroid after the administration of 5 mCi Tc-99m sestamibi shows diffuse uptake throughout the thyroid, with a focal area of increased uptake (arrow) in the lower pole of the right lobe.



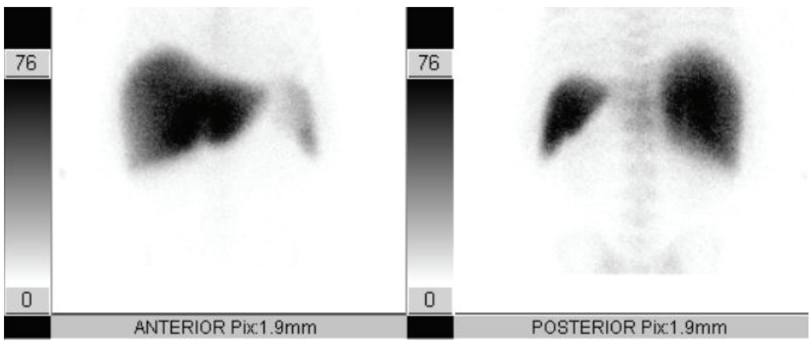
Delayed anterior pinhole image shows interval partial washout of Tc-99m sestamibi. There is a persistent focus of radiotracer uptake (arrow) in the right lower pole. A right inferior parathyroid adenoma was confirmed at surgery.

- Dual-phase (early and delayed) imaging with Tc-99m sestamibi can localize a suspected parathyroid adenoma in a patient with elevated parathyroid hormone.
- A parathyroid adenoma shows increased uptake on early images and persistent retained activity on delayed images. In contrast, a thyroid adenoma will also initially show increased uptake but will then wash out on delayed images.
- Parathyroid tissue does *not* take up Tc-99m pertechnetate, which can be administered in indeterminate cases.
- SPECT/CT imaging is also frequently performed to provide more precise localization of uptake to aid in presurgical planning.

LIVER-SPLEEN IMAGING

Radiotracer: Tc-99m sulfur colloid

- Sulfur colloid is taken up by reticuloendothelial cells, which are found in the liver, spleen, and bone marrow. Sulfur colloid is also taken up by Kupffer cells in the liver. Hepatic Kupffer cells make up only approximately 10% of the liver mass.
- 80–90% of sulfur colloid particles are taken up by the liver. Most of the remainder is taken up by the spleen, and a small amount taken up in the bone marrow. The bone marrow is not normally seen at typical windowing levels.
- Although Tc-99m has a physical half-life of 6 hours, sulfur colloid is rapidly cleared with a biologic half-life of 2–3 minutes.



Normal liver-spleen scan following administration of 4.2 mCi Tc-99m sulfur colloid shows homogeneous uptake in the liver and spleen. Note splenic uptake is slightly lower in intensity than the liver. Bone marrow uptake is faint.

Focal decreased hepatic uptake on sulfur colloid scan

- The most common cause of a photopenic defect (complete absence of radiotracer) is a hepatic cyst. It may be difficult to distinguish focal *decreased* uptake from a *photopenic* defect if the lesion is small.
- Most hepatic masses cause focal decreased radiocolloid uptake, including hepatocellular carcinoma (HCC), adenoma, and abscess. Focal decreased uptake should raise concern for HCC in a patient with any risk factors for HCC such as cirrhosis or chronic hepatitis.

Focal increased hepatic uptake on sulfur colloid scan

- Focal nodular hyperplasia, discussed below, may hyperconcentrate radiocolloid.
- A regenerating nodule in a cirrhotic liver can cause focal increased radiocolloid uptake.
- Budd-Chiari syndrome, or hepatic vein thrombosis, can lead to increased uptake in the caudate lobe in the later stage of disease.

Colloid shift

- Colloid shift is increased sulfur colloid accumulation within the spleen and bone marrow. It suggests liver dysfunction, most commonly due to cirrhosis.

Diffuse pulmonary uptake

- Diffuse pulmonary uptake on a sulfur colloid scan is nonspecific, and can be seen in:

Cirrhosis.

COPD with superimposed infection.

Langerhans cell histiocytosis.

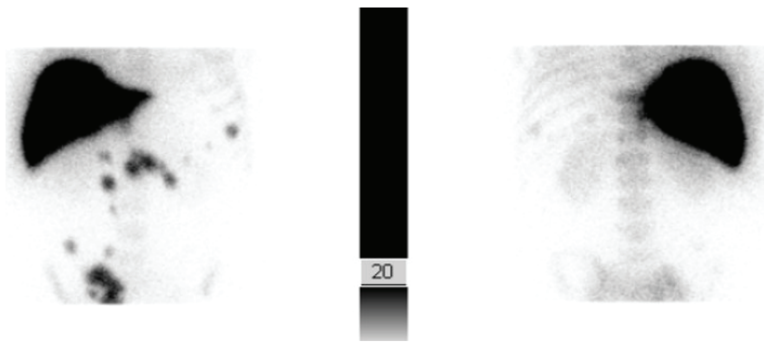
High serum aluminum (either due to antacids or excess aluminum in the colloid preparation).

Focal nodular hyperplasia (FNH)

- Focal nodular hyperplasia (FNH) is a benign liver mass that can have a variable appearance on a sulfur colloid scan. Most commonly, FNH will be indistinguishable from background liver due to Kupffer cells within the FNH. FNH may also have increased uptake due to a combination of hypervascularity and Kupffer sulfur colloid uptake. In approximately 1/3 of cases there is insufficient colloid concentration and FNH may appear as a photopenic defect.
- In contrast to FNH, a hepatic adenoma does not contain Kupffer cells and will be consistently photopenic.
- An adjunct or alternative to sulfur colloid scan is a HIDA scan. FNH contains biliary ductules so it should demonstrate radiotracer uptake on a HIDA scan.

Intrapancreatic splenic tissue

- The presence of an intrapancreatic splenule can be confirmed by sulfur colloid scan in indeterminate cases, where the suspected tissue will show uptake. A Tc-99m damaged red cell study can also be performed.



Splenosis: Liver-spleen scan (5.5 mCi Tc-99m sulfur colloid administered) of a patient with history of splenectomy following trauma.

Anterior and posterior images show scattered focal uptake in the abdomen correlating to intraperitoneal soft tissue masses on SPECT/CT (not shown), consistent with splenosis.

Esophageal and gastric motility

Radiotracer: Tc-99m sulfur colloid or DTPA

- Sulfur colloid or DTPA are mixed with food and ingested to assess gastroesophageal motility.

Esophageal transit

- After a 6-hour or overnight fast, patient is positioned supine and instructed to swallow 10 mL water containing Tc-99m sulfur colloid or DTPA, then attempt dry swallows to clear normal residual activity in the esophagus. This process may be repeated several times.
- Normal esophageal transit time for water is 5–11 seconds, with more than 90% of peak activity clearing the esophagus after 15 seconds. Reduced transit time is seen in patients with scleroderma and achalasia.

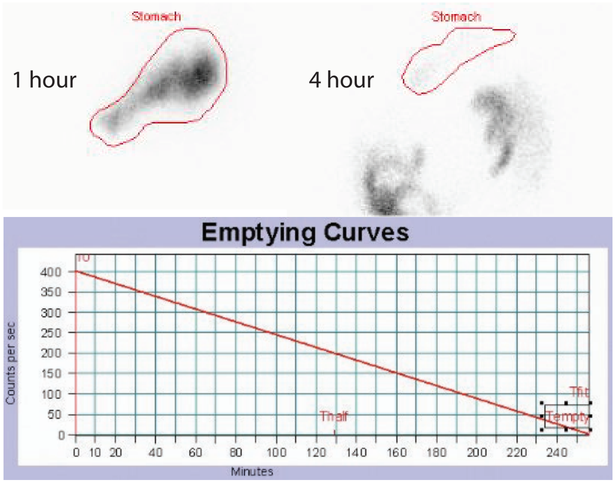
Gastroesophageal reflux

- Scintigraphy for gastroesophageal reflux is most commonly performed in neonates. After an overnight fast, the infant is fed formula or milk mixed with Tc-99m sulfur colloid. Sequential images are acquired to assess for reflux of gastric content into the esophagus. Delayed images of the lungs can be obtained to evaluate for aspiration.
- A similar protocol is used for older children and adults, but using orange juice.

Salivagram for aspiration

- Salivagram is more sensitive in detecting aspiration in pediatric patients compared to reflux scintigraphy. A drop of Tc-99m sulfur colloid is placed on the patient’s tongue base or sublingual region to allow mixture with saliva. Rapid images are then obtained to assess for activity in the bronchi and lungs.

Gastric emptying



Normal gastric emptying scan:

1 mCi Tc-99m sulfur colloid mixed in egg whites was administered orally with white bread, jelly and water. Patient consumed 100% of the solid portion. Sample static images at 1 hour and 4 hours show normal gastric retention (1% at 4 hours). The emptying curve shows counts graphed to time.

- After a 6-hour or overnight fast, the patient is given a standardized solid meal containing up to 1 mCi Tc-99m sulfur colloid mixed in scrambled egg whites, along with toast, jam and water. Patient should ingest the entire meal within 10 minutes. Subsequent images of the stomach are then acquired every 60 minutes up to 4 hours.
- Normal residual activity in the stomach at 4 hours is <10%.
- An alternative liquid phase gastric emptying study can be performed using water containing Tc-99m DTPA, with a shorter imaging time because liquids normally empty more rapidly.

Meckel imaging

Radiotracer: Tc-99m pertechnetate

- Technetium-99m pertechnetate localizes to gastric mucosa.

Meckel diverticulum

- Meckel diverticulum is a remnant of the embryological omphalomesenteric duct, most commonly located in the distal ileum. Approximately 10–60% of Meckel diverticula contain ectopic gastric mucosa, which may result in mucosal damage and GI bleeding.
- A positive Meckel scan demonstrates a focal area of increased activity, typically in the right lower quadrant. A lateral view is often helpful to ensure that activity is anterior and not associated with the ureter.
- Other causes of uptake in the right lower quadrant include appendicitis and intussusception that can cause more diffuse, regional increased activity due to hyperemia.
- Sensitivity of Meckel scan can be increased by a number of medications:
  - Pentagastrin increases uptake of pertechnetate by gastric mucosa.
  - H2 blockers such as cimetidine block release of pertechnetate by the mucosal cells.
  - Glucagon decreases intestinal peristalsis, thereby slows washout of pertechnetate from the diverticulum.

Gastrointestinal bleeding

Radiotracer: Tc-99m labeled RBCs

- Technetium-99m labeled red blood cells are prepared in-vitro by mixing 1–3 mL of anticoagulated blood with stannous chloride and an oxidizing agent; Tc-99m is then added. The labeling efficiency is 95%. The labeling procedure takes more than 20 minutes.
  - An in-vivo technique provides much noisier images due to worse labeling efficiency and resultant free pertechnetate, and is therefore uncommonly performed.
- Alternatively, Tc-99m sulfur colloid can be used. Sulfur colloid does not require significant preparation time, but has rapid blood clearance with a vascular half-life of 2–3 minutes.

Acute GI bleed

- A tagged red blood cell study can be used to identify patients who may be suitable for angiography versus those who are not. Bleeding rates as low as 0.05–0.1 mL/min can be detected, compared to 1 mL/min for angiography.
- A positive study shows an initial focus of activity that changes shape or position over time due to antegrade or retrograde peristalsis of intraluminal blood.

GI bleed: Sequential images of the lower abdomen on Tc-99m tagged RBC study show a focus of radiotracer uptake in the right lower quadrant, in the expected distribution of the superior mesenteric artery, which changes shape and position over time (arrows).

Superior mesenteric artery angiogram in the same patient performed within one hour of the above tagged RBC study shows a focus of contrast extravasation in a distal ileocolic branch (arrow).

This patient was treated with coil embolization of the distal ileocolic branch artery.

Hepatobiliary imaging (HIDA)

Radiotracers: Tc-99m IDA agents

- Tc-99m-iminodiacetic acid (IDA) analogs are used to image the biliary system. Regardless of the tracer, the test is typically called a "HIDA" scan.
- **Disofenin** allows visualization of the biliary system with bilirubin levels as high as 20 mg/dL and has 90% hepatic uptake. **Mebrofenin** allows visualization of the biliary system with bilirubin levels as high as 30 mg/dL and has an even higher 98% hepatic uptake. Both are actively transported into hepatocytes but are not conjugated.
- Critical organ is the gallbladder wall.

HIDA protocol

- Patient should be NPO for 6 hours prior, but must have eaten within 24 hours. If the patient has been NPO for >24 hours, **cholecystokinin** (CCK; dose 0.02 μg/kg, given as slow infusion) is given to empty the gallbladder before radiotracer is administered.

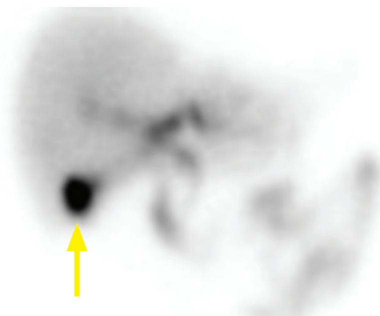
  CCK must be administered slowly or the patient may experience an exacerbation of their symptoms.

- Dynamic imaging of the right upper quadrant begins immediately after injection of 5 mCi Tc-99m IDA. As soon as the gallbladder is visualized, acute cholecystitis is ruled out.
- If the gallbladder is not visualized by one hour, **morphine** is given (0.04 mg/kg, up to a maximum dose of 4 mg) and imaging continues for 30 more minutes. Morphine contracts the sphincter of Oddi, redirecting bile into the cystic duct.

  Morphine should only be given if tracer is visualized in the small bowel, otherwise there is the theoretical risk of worsening a potential common bile duct obstruction. However, nonvisualization of tracer in the small bowel is not specific for common bile duct obstruction.
- If the patient has a morphine allergy, an alternative is to image for a total of 4 hours.

Normal HIDA scan

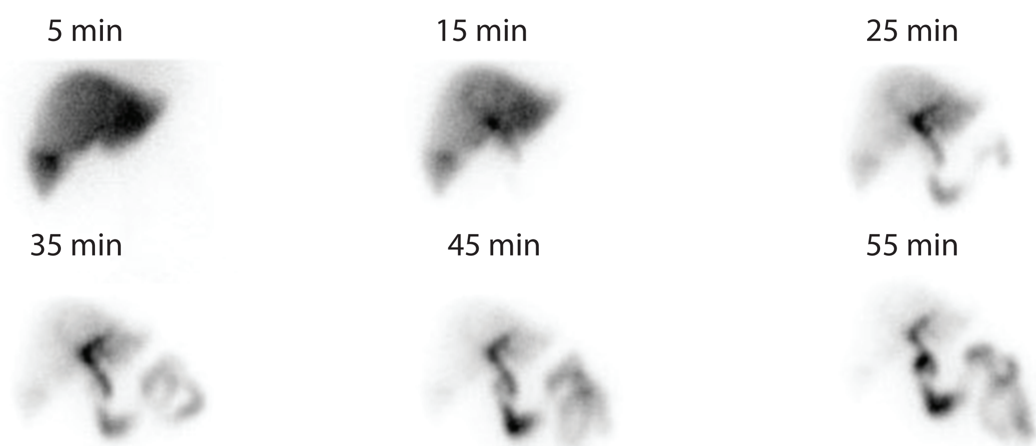
- In a normal HIDA scan, the liver is visible by 5 minutes.
- The gallbladder is typically seen by 15 minutes due to radiotracer flow into the cystic duct.
- Tracer should be seen in the small bowel to ensure a patent common bile duct.
- Small amount of bile reflux into the stomach can be a normal finding.



Single image from a HIDA scan 15 minutes after 4.1 mCi Tc-99m disofenin administered shows normal visualization of the liver, gallbladder (arrow), and small bowel.

Acute cholecystitis

- Almost all patients with acute cholecystitis have cystic duct obstruction.
- Nonvisualization of the gallbladder after 90 minutes of imaging and morphine augmentation is approximately 86–98% sensitive for the diagnosis of acute cholecystitis, with a false-positive rate of approximately 6%, most commonly due to pancreatitis and biliary stasis.
- The *rim sign* describes increased hepatic activity surrounding the gallbladder fossa, thought to be due to hyperemia and is an ancillary finding of acute cholecystitis.
- The *cystic duct sign* describes a small focus of activity in the cystic duct just proximal to the site of obstruction.



Acute cholecystitis: Sequential images from a HIDA scan with 3.87 mCi Tc-99m disofenin administered shows prompt visualization of the liver. The small bowel is visualized by 25 minutes. There is no visualization of the gallbladder during the course of the exam. The patient had a morphine allergy so 4-hour delayed images were obtained instead (not shown), which also do not demonstrate the gallbladder.

Acute cholecystitis (continued)

- False positive HIDA (gallbladder nonvisualization *without* acute cholecystitis) can be due to:

Recent meal (within 4 hours) or prolonged fasting (greater than 24 hours). Administration of CCK *immediately* prior to the exam, which can cause persistent sphincter of Oddi relaxation. Total parenteral nutrition.	Pancreatitis. Severe illness. Chronic cholecystitis. Cholangiocarcinoma of the cystic duct (very rare).

- False negative HIDA (gallbladder visualization *with* acute cholecystitis) is very rare and can be due to:

Acalculous cholecystitis with a patent cystic duct.
Duodenal diverticulum simulating the gallbladder; however, a lateral view would differentiate.
Choledochal cyst simulating the gallbladder.
Right extrarenal pelvis in jaundiced patients with increased renal excretion of tracer.

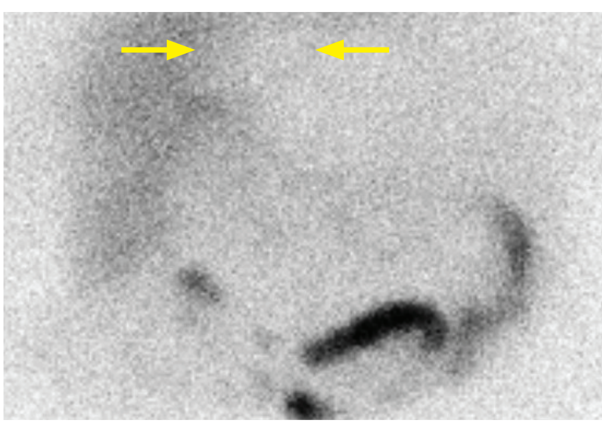
Chronic cholecystitis

- Chronic cholecystitis is long-standing gallbladder inflammation causing loss of normal gallbladder function, predisposing to formation of stones.
- Chronic cholecystitis may be a cause of chronic recurrent abdominal pain.
- Diagnosis can be difficult as there is no single pattern that is pathognomonic of chronic cholecystitis based on HIDA. In fact, most cases of chronic cholecystitis have a normal HIDA.
- A low **gallbladder ejection fraction** (GBEF) is thought to be suggestive of chronic cholecystitis. To measure the ejection fraction, pre- and post-CCK injection gallbladder counts are compared. A GBEF <35% suggests chronic cholecystitis, although the reliability of this finding is controversial as ejection fraction is dependent on the rate of CCK injection and standard infusion protocols have not been determined.

Biliary obstruction

- Acute high-grade obstruction of the common bile duct can result in the *liver scan sign*, where there is uptake in the liver but no visualization of the biliary tree or small bowel. Intrahepatic cholestasis or hepatitis can have a similar appearance.
- Delayed clearance of activity from the common bile duct suggests partial obstruction.

Hepatic dysfunction

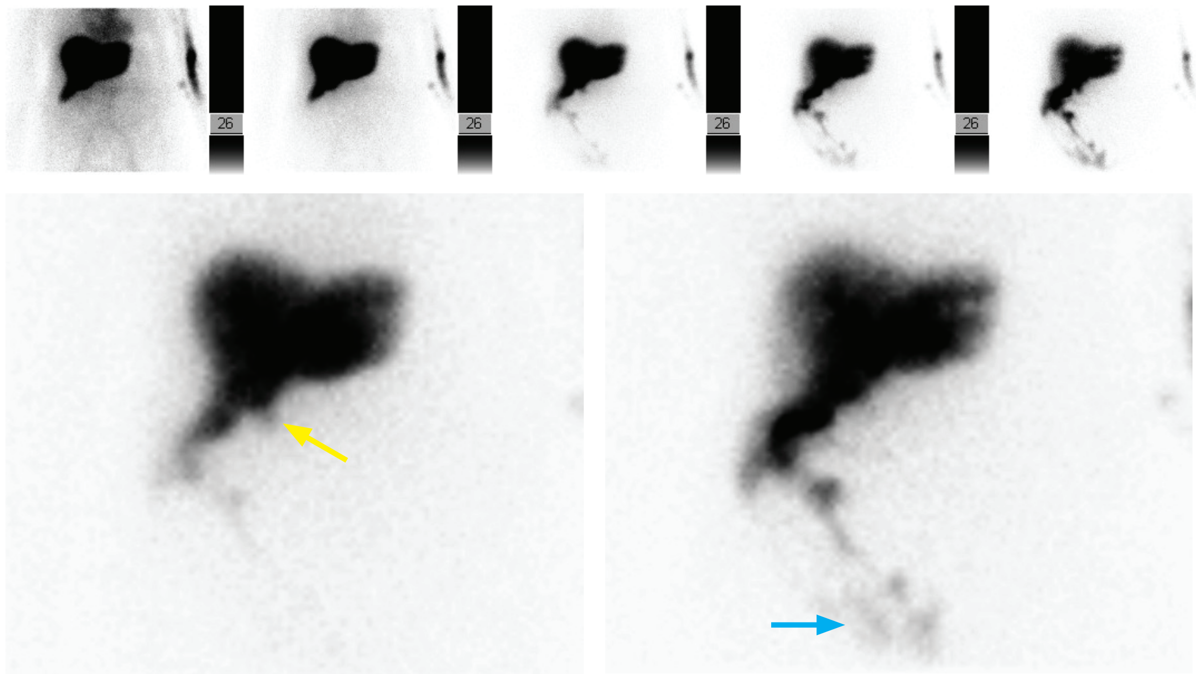


Severe hepatic dysfunction in a patient with cholangiocarcinoma:

Single image from a HIDA obtained 30 minutes after 6.1 mCi Tc-99m disofenin administered shows a large amount of background and blood pool uptake with minimal tracer in the liver. A large photopenic defect in the left lobe of the liver (arrows) corresponds to the patient's cholangiocarcinoma. There is small bowel tracer excretion.

- Since the IDA tracers are actively transported into hepatocytes and then secreted into the bile, severe hepatic dysfunction will cause very poor hepatic uptake and blood pool clearance will be delayed.
- Since functioning hepatocytes are necessary to extract the tracer, a hepatic mass will appear as a focal photopenic defect.

Biliary leak



Bile leak: Sequential images (top row) and two enlarged still images (bottom two images) from a HIDA scan with 5.4 mCi Tc-99m disofenin administered to a patient with right upper quadrant pain status post cholecystectomy shows tracer accumulating in the gallbladder fossa (yellow arrow) and pooling in the right paracolic gutter (blue arrow). The gallbladder is surgically absent. The leak was thought to arise from the cystic duct stump.

- A HIDA scan is an accurate method to assess for the presence of a biliary leak. Imaging can be performed in the right lateral decubitus position to promote dependent pooling of bile.

Pulmonary

Radiotracers

Tc-99m MAA (perfusion)

- Technetium-99m macro-aggregated albumin (MAA) is a particulate radiopharmaceutical that lodges in the pulmonary capillary bed and can therefore be used to evaluate pulmonary perfusion. Most particles are between 10–30 µm in size.
- Typically 3–5 mCi Tc-99m MAA are administered intravenously, comprising between 200,000 and 600,000 particles. Particles begin to break down in approximately 30 minutes. In children, pregnant patients, patients with mild pulmonary hypertension, and patients with a known right-to-left shunt, the dose can be halved to approximately 100,000 particles.
- A relative contraindication to MAA is severe pulmonary hypertension, as obstruction of even a few pulmonary capillaries can cause clinical worsening.
- A **right-to-left shunt** causes immediate renal and brain uptake after intravenous injection. A Tc-99m MAA study may be able to quantify the shunt fraction in these patients. Note that renal uptake can also be seen if free pertechnetate is present, but evaluation of the head and neck can differentiate free pertechnetate from a right-to-left shunt. Free pertechnetate is taken up by the thyroid, but not the brain. A right-to-left shunt, in contrast, demonstrates immediate brain uptake and no significant uptake in the neck.
- Clumping of MAA can be seen when MAA is inadvertently drawn back into injection syringe, causing coagulation with the patient's blood.

Xenon-133 (ventilation)

- Xenon-133 is an inhaled gas with a physical half-life of 5.3 days, which emits 81 keV gamma photons and is also a beta-emitter. The critical organ is the trachea. The biologic half-life is very short because the vast majority of the gas is exhaled. Although 10–20 mCi is administered, there is very little radiation exposure.
- Xenon-133 is imaged posteriorly to avoid breast artifacts, as the relatively low keV is easily attenuated by soft tissue. Wash-in/washout imaging can be performed to evaluate for air trapping, which is seen in COPD.
- Xenon-133 requires good patient cooperation as the patient must breathe on a closed spirometer for several minutes. Exhaled xenon-133 must be carefully disposed of, either exhausted to the atmosphere or trapped in a charcoal trap until it decays. Xenon-133 must be administered in a negative pressure room to prevent accidental leakage.
- Because xenon-133 is soluble in fat, it can accumulate in the liver, particularly a fatty liver.

Tc-99m DTPA (ventilation)

- Tc-99m DTPA is a technetium-labeled aerosol. Unlike Xenon-133, DTPA does not allow for dynamic wash-in/washout imaging. Once inhaled, the particles remain in place for approximately 45–60 minutes (20 minutes in smokers). 30 mCi is typically administered.
- Compared to xenon, Tc99m DTPA has greater ease of use. Specific advantages include the ability to image in multiple projections, little requirement for patient cooperation, no need for exhaust systems, ability to use portably and be delivered through mechanical ventilation.
- Swallowed activity can be seen in the esophagus and stomach.

Clinical V/Q Scanning in the era of CT pulmonary angiography

Diagnosis of pulmonary embolism in pregnancy

- Pulmonary embolism is the leading cause of death in pregnancy in the developed world, and diagnosis must be weighed against radiation exposure to the mother and the fetus.

> CT pulmonary angiogram (CTPA): Maternal breast dose approximately 10–70 mGy (much greater than ACR mammography guidelines of 3 mGy/breast). Fetal dose 0.01–0.66 mGy.
>
> V/Q scan: Maternal breast dose <1.5 mGy. Fetal dose 0.5–1.1 mGy (higher than CTPA but still considerably lower than the 100 mSv threshold for potential teratogenic effects).
>
> Low-dose perfusion (Q) scan: Maternal breast dose 0.6 mGy. Fetal dose 0.1–0.2 mGy.

- In most cases, only perfusion scan is necessary and ventilation scan is not commonly performed. The administered dose of Tc-99m MAA is typically one-half of the standard dose, or approximately 100,000 particles.
- Diagnostic accuracy for PE has been shown to be equivalent between CTPA and Q scan.
- The primary advantage of CTPA is the demonstration of an alternative diagnosis not seen on plain radiographs approximately 5% of the time and a lower fetal dose, at the expense of increased breast radiation exposure.
- Nuclear medicine perfusion scan should be reserved for suspected pulmonary embolism in a pregnant patient with a normal chest radiograph, no history of asthma or COPD, and a negative duplex ultrasound of lower extremities. In clinical practice, CTPA is more readily available and often the prefered modality.

Diagnosis of pulmonary embolism with PIOPED II

High probability: Specificity for pulmonary embolism 97%, positive predictive value 88%

High probability for pulmonary embolism: RPO perfusion images (same image, without and with annotations) after 4.7 mCi Tc-99m MAA administered show two large segmental defects in the posterior basal (red) and anterior basal (brown) segments of the right lower lobe. The ventilation scan (not shown) was normal.

- Two or more large mismatched segmental defects without associated radiographic abnormality denote high probability for pulmonary embolism.

Intermediate probability: Not clinically helpful, further imaging is required

- One large segmental mismatched perfusion defect denotes intermediate probability.
- A triple matched defect in the *lower lung* is considered an intermediate probability. A triple matched defect describes a defect on perfusion, a matched defect on ventilation, and a corresponding abnormality on the chest radiograph.

Low probability: Negative predictive value 84%

- A single large or moderate matched V/Q defect is low probability of pulmonary embolism.
- Other low probability results are absent perfusion of an entire lung, or more than three small segmental lesions.

Very low probability

- Nonsegmental defects denote very low probability.
- The *stripe sign* peripheral to a perfusion defect denotes very low probability for pulmonary embolism. The *stripe sign* represents a thin line of MAA uptake between a perfusion defect and the adjacent pleural surface, representing intervening perfused lung.
- A solitary triple-matched defect in the mid-to-upper lung denotes very low probability. In contrast, the previously mentioned triple-matched defect in the lower lung is more likely to represent PE and is considered an intermediate probability finding.

Normal

- If the perfusion scan is normal, the exam is normal and the ventilation study is not required.

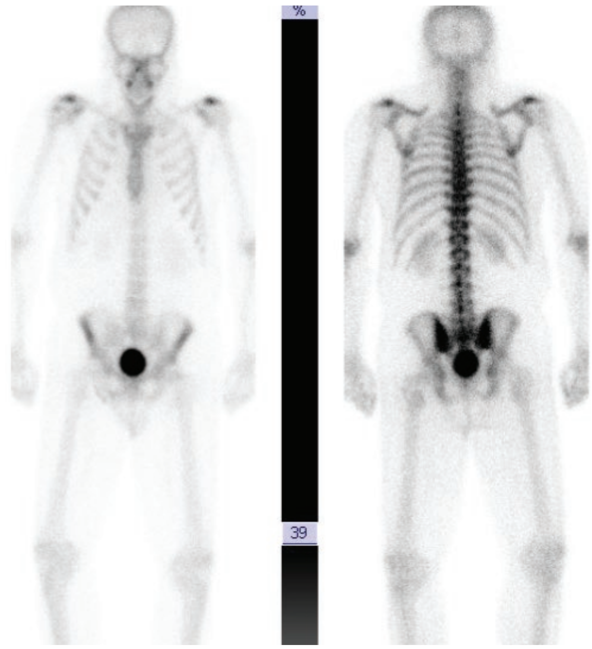
Musculoskeletal

Radiotracers

Tc-99m MDP or HDP

- Tc-99m MDP and HDP are technetium-labeled diphosphonates which deposit via chemisorption in the mineral phase of bone. 20 mCi is typically administered, with imaging performed 2–4 hours later.
- Rapid renal excretion is normal; bladder is the critical organ. Patients should void before imaging to prevent the bladder from obscuring a pelvic lesion.
- A three-phase study incudes:

> 1) Radionuclide **angiogram (flow)** evaluates blood flow, with images taken every few seconds. Increased flow suggests hyperemia.
>
> 2) **Blood pool** evaluates the extracellular distribution immediately following the blood flow phase.
>
> 3) Standard **delayed** (skeletal) images are performed approximately 3 hours after injection.



Normal Tc-99m MDP bone scan in anterior and posterior projections, with 20 mCi administered. Note the normal renal and bladder uptake.

Common patterns and clinical applications

Patterns with low probability of metastatic disease (in a patient with a known malignancy)

- A single focus of uptake in a rib is thought to represent malignancy only ~10% of the time.
- Uptake in similar locations in two or more adjacent ribs is almost always due to trauma.
- Multiple adjacent photopenic bony lesions are unlikely to be metastases, but may represent infarction, avascular necrosis, or sequela of radiation therapy.

Patterns with high probability of metastatic disease

- A single sternal lesion in a patient with breast cancer is due to metastasis ~80% of the time.
- Multifocal areas of increased activity in nonadjacent ribs are suspicious for metastases.
- A single photopenic lesion in patients with a known malignancy (especially neuroblastoma, renal cell carcinoma, and thyroid cancer) is due to metastasis 80% of the time.
- Beware of the *flare phenomenon* within three months of starting chemotherapy, where treated bone metastases demonstrate increased uptake due to healing. Tracer uptake should regress by 6 months following treatment.

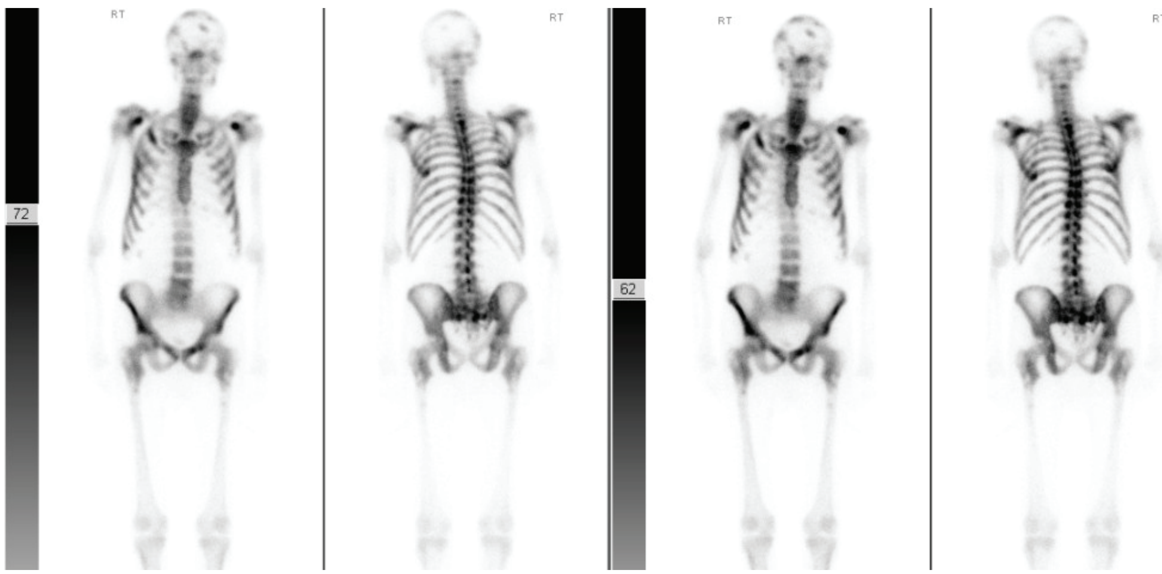
Cancers likely to produce a false-negative bone scan

- While bone scan is highly sensitive for osseous metastases that induce bone turnover or are predominantly osteoblastic, such as prostate cancer, it has low sensitivity for predominantly osteoclastic or lytic metastases. These tumors include multiple myeloma, renal cell carcinoma, thyroid carcinoma, aggressive anaplastic tumors, and neuroblastoma.
- Marrow-replacing tumors such as lymphoma can also produce negative bone scans.
- Aggressive and lytic bony lesions or those confined to the marrow are better seen on F-18 FDG PET/CT.

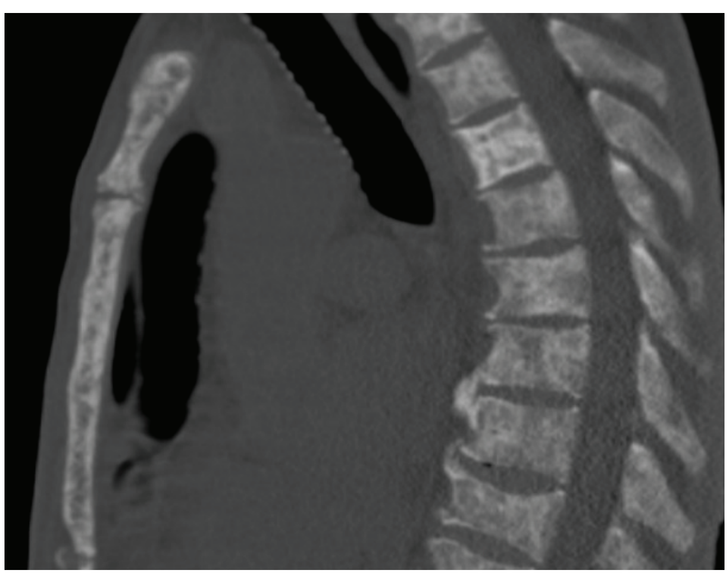
Increased soft tissue uptake

- Diffuse soft tissue uptake can be seen in renal failure.
- Increased uptake in the brain, heart, or spleen may be due to recent infarction.
- Malignant pleural effusion or ascites can cause increased uptake.
- Soft-tissue metastases containing calcium can have increased uptake, including osteosarcoma, neuroblastoma, or mucin-producing tumors (gastrointestinal and ovarian).
- Soft tissue uptake associated with dystrophic calcifications can be seen after trauma or in inflammatory diseases such as myositis ossificans, dermatomyositis, or rhabdomyolysis.
- In the breasts, faint uptake can be normal, but focal intense uptake suggests breast carcinoma. Tracer uptake can be seen at site of recent breast procedure (e.g., biopsy).
- Diffuse parenchymal uptake in kidneys can be seen with urinary obstruction, hyperparathyroidism, chemotherapy, or thalassemia.

Superscan



Superscan: Anterior and posterior whole body bone scan at two different windowing levels after 21 mCi Tc-99m MDP administered shows diffusely increased skeletal uptake in the axial skeleton and proximal femurs. There are scattered foci of uptake in the calvarium. No renal uptake is seen.



Sagittal CT in the same patient shows diffuse sclerotic appearance of the entire skeleton. This patient had metastatic prostate carcinoma.

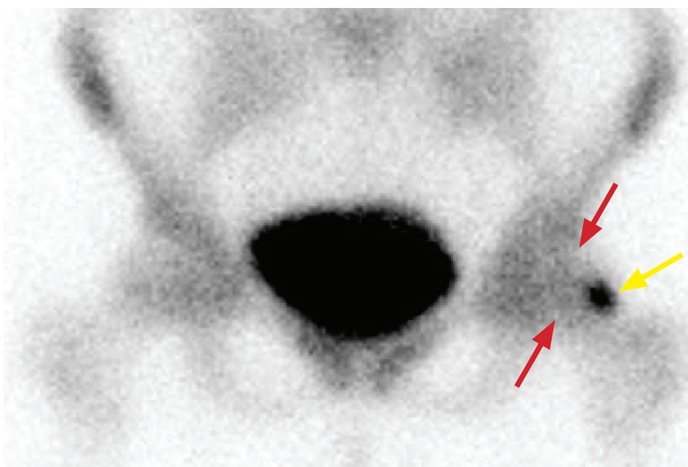
- A *superscan* represents diffusely increased osseous uptake. Sometimes the uptake can be so diffuse that it may look normal on first glance. The clue to the presence of a superscan will be nonvisualization (or very faint visualization) of the kidneys.

Superscan (continued)

- A superscan is most commonly due to metastatic prostate cancer. Other malignancies producing a superscan include breast cancer and lymphoma.
- A superscan can also be caused by metabolic bone disease in hyperparathyroidism and renal osteodystrophy. If the cause for the superscan is metabolic, then the entire long bones tend to demonstrate diffusely increased uptake. In contrast, with metastatic disease the axial skeleton and proximal humeri and femora are primarily affected.

Primary bone tumors

- **Osteosarcoma** typically causes markedly increased uptake, often with increased uptake in the entire affected limb.
- **Ewing sarcoma** features intense and homogeneous activity. It may be positive on all three phases of a three-phase bone scan, mimicking osteomyelitis.
- Benign bone tumors that have intense uptake include osteoid osteoma, osteoblastoma, fibrous dysplasia, giant cell tumor, and aneurysmal bone cyst.
- **Osteoid osteoma** features a vascular central nidus which demonstrates intense activity. The *double density sign* describes an intense focus of uptake corresponding to the nidus, surrounded by relatively increased uptake representing hyperemia. The differential of the *double density sign* includes osteoid osteoma, Brodie abscess, less likely stress fracture.



Double density sign of osteoid osteoma: Tc-99m MDP bone scan (25.9 mCi administered) shows asymmetric regional uptake in the left femoral head (red arrows), with a focus of intense uptake at the left lateral head/neck junction (yellow arrow).

Case courtesy Roger Han, MD, Brigham and Women's Hospital.

Coronal CT of the left hip shows a well circumscribed lucent lesion (arrow) with a sclerotic rim and lucent nidus, with mild surrounding osteopenia, consistent with osteoid osteoma.

Fracture

- An **acute** fracture (from the initial injury up to 3–4 weeks) will show increased radiotracer uptake surrounding the fracture site. 95% of fractures are positive after one day in patients under 65 years old. Note that skull fractures rarely show activity.
- In the **subacute** phase (until two to three months) radiotracer activity becomes more focal at the fracture.
- In the **healing** phase (variable time course, with ~40% remaining abnormal after one year) there is a gradual decrease in radiotracer activity.
- **Toddler's fracture** represents a spiral fracture of the tibia and occurs in recently ambulatory children. Skeletal scintigraphy is sensitive and specific, and routinely used in workup of a limping child if the radiographs are negative.

Stress fracture

- A stress fracture represents a fracture due to abnormal stress in a normal bone.
- Common sites of stress fractures include the tibial diaphysis, the femoral neck, and the metatarsals. Metatarsal stress fractures are the most common stress injury in the foot/ankle, with 90% occurring in the second or third metatarsals.
- The bone scan is typically positive by the time the patient has pain. A stress fracture is positive on all three phases of the bone scan.

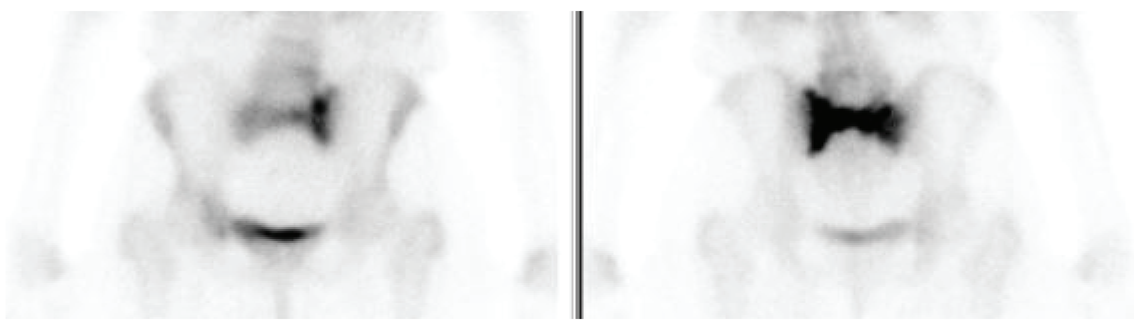
Shin splints (medial tibial stress syndrome)

- Shin splints cause pain from periostitis at the tibial insertions of the anterior tibialis and soleus muscles.
- A three-phase bone scan shows *normal* blood flow and blood pool images, with linearly increased uptake in the tibia on delayed (skeletal) phase.

Anterior projection Tc-99m MDP bone scan (25 mCi administered) shows linear uptake along the anterior cortex of the right tibia (arrow), consistent with shin splint. Degenerative-type uptake is seen at both knees.

Insufficiency fracture

- An insufficiency fracture represents a fracture in response to normal stress in an abnormal bone due to underlying osteoporosis.
- A sacral insufficiency fracture typically shows "H"-shaped uptake in the sacrum, often referred to as the *Honda sign*.
- Insufficiency fracture of the knee is a cause of atraumatic knee pain in the elderly. It typically appears as intense uptake in the medial femoral condyle. This condition was previously called spontaneous osteonecrosis of the knee (SONK) but has seen been discredited as it is insufficiency rather than osteonecrosis.



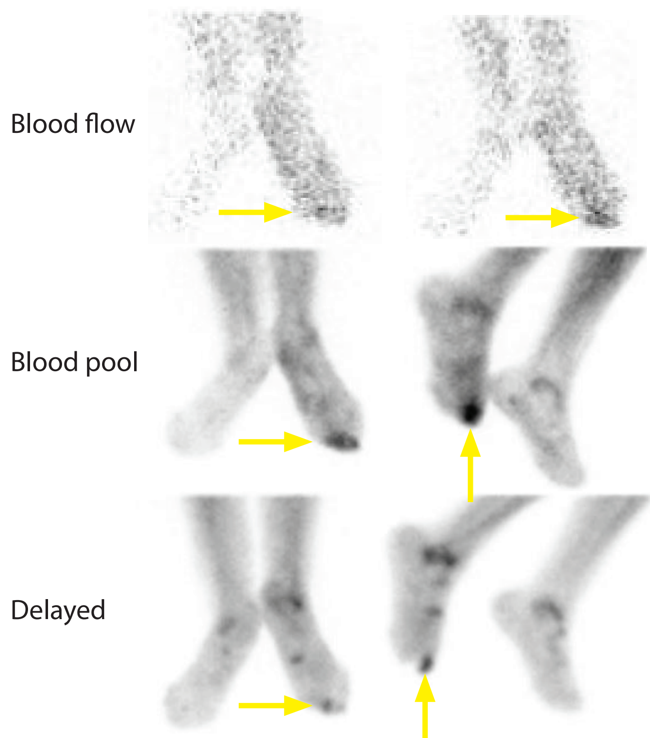
Anterior and posterior projections from a Tc-99m MDP bone scan (25 mCi administered) show uptake in the sacrum with a H-shaped configuration, representing the *Honda sign* of sacral insufficiency fracture.

Prosthesis evaluation

- Evaluation of a painful hip or knee prosthesis for loosening or infection is a common indication for a bone scan.
- In a cemented prosthesis, it is normal to see activity surrounding the prosthesis up to 12 months. In a non-cemented prosthesis, activity may remain increased up to two years as bony ingrowth continues.
- In evaluation of a hip prosthesis, *focal* activity at the lesser trochanter (which acts as a fulcrum site) and distal femoral prosthetic tip seen >1 year for cemented or >2 years for non-cemented prosthesis suggests loosening. In contrast, *generalized* increase in radiotracer activity surrounding the prosthesis may suggest osteomyelitis.
- Mild to moderate activity that is limited to the greater trochanter/intertrochanteric region is often due to heterotopic ossification.

Osteomyelitis and musculoskeletal infection

- Osteomyelitis is positive on all three phases of a bone scan. A positive three-phase bone scan is very specific for osteomyelitis in the presence of a normal radiograph (confirming that no fracture is present).
- To evaluate for osteomyelitis in the presence of an underlying abnormality, such as a fracture or prosthesis, *WBC imaging* (indium-111 or Tc-99m labeled WBCs) combined with a Tc-99m sulfur colloid *marrow scan* can increase specificity for osteomyelitis. Focal WBC activity in a region devoid of colloid marrow activity is suggestive of osteomyelitis.
- Although rarely performed, *gallium-67 scan* can also increase specificity for osteomyelitis if the gallium uptake exceeds the bone scan uptake in the area of concern.
- **Cellulitis** shows increased MDP activity during the blood flow and soft tissue phases, but the skeletal phase is negative.
- **Septic arthritis** is positive on all three phases of the bone scan on *both* sides of the joint.
- **Discitis** shows increased skeletal activity in two adjacent vertebral bodies.



Three-phase bone scan following administration of 26.2 mCi Tc-99m MDP in patient with nonhealing ulcer of the left great toe:

Focal increased uptake (arrows) is present at the left great toe on blood flow, blood pool, and delayed phases, suggesting osteomyelitis.

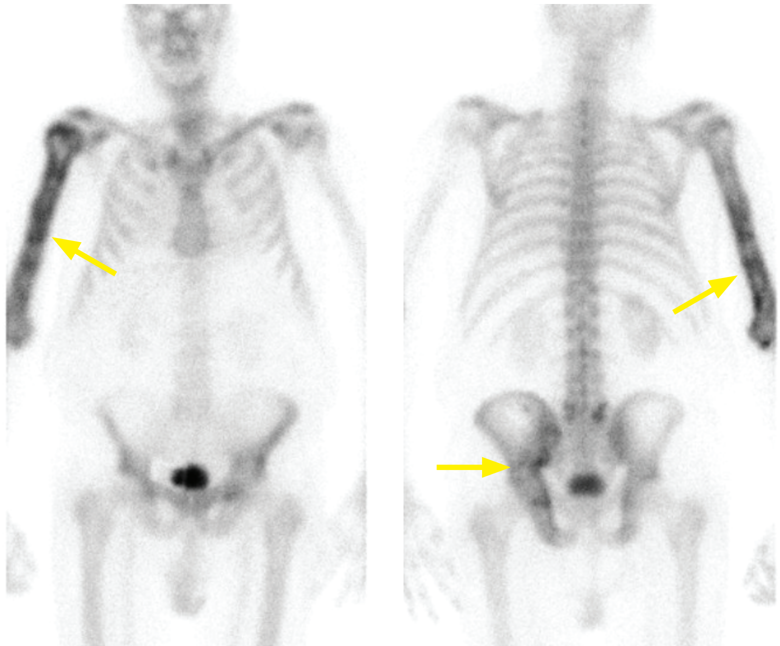
Hypertrophic pulmonary osteoarthropathy

- Hypertrophic pulmonary osteoarthropathy is long bone diaphyseal periosteal reaction most commonly due to pulmonary disease, such as lung cancer.
- Bone scan shows increased parallel lines of activity along the cortex of long bones.

Avascular necrosis (AVN)

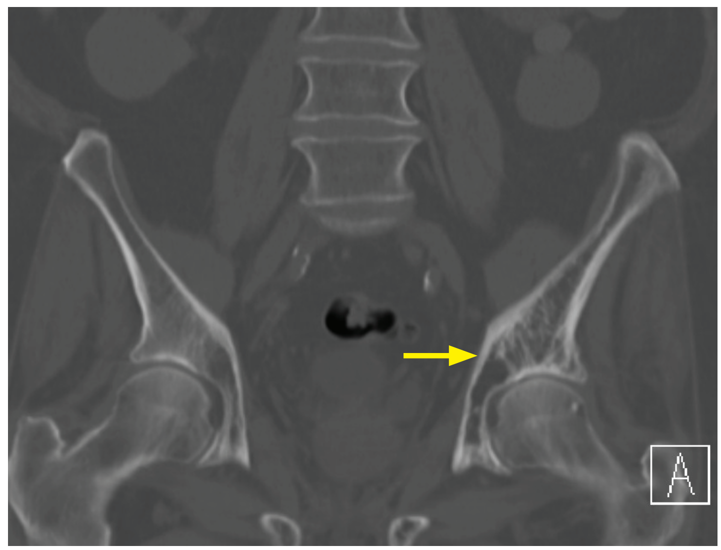
- Avascular necrosis (AVN) initially shows decreased radiotracer activity in the affected region, followed by a hyperemic phase with increased uptake.
- SPECT imaging of AVN (especially of the hips) will often show a rim of increased uptake with central photopenia, thought to represent revascularization progressing from outside in.

Paget disease



Paget disease: Anterior (left image) and posterior projections from a Tc-99m MDP bone scan (20 mCi administered) show heterogeneously increased radiotracer uptake involving the right humerus, and to a lesser extent the left hemipelvis, best seen on the posterior projection (arrows).

Radiograph of the right humerus shows cortical and trabecular thickening.



Coronal CT of the pelvis shows asymmetric coarsened trabecular thickening of the left ilium (arrow).

- Paget disease is an idiopathic disturbance of osteoclastic and osteoblastic regulation. Paget has three discrete phases including lytic, mixed, and sclerotic phases. The lytic (early) phase is typically positive on bone scan and often negative on radiography. The mixed phase is abnormal on bone scan and radiography. The sclerotic phase shows persistent radiographic changes, but the bone scan activity may subside.
- One complication of Paget disease is malignant degeneration to osteosarcoma. A focal cold lesion should raise concern for necrosis in a region of malignant degeneration, although this may be best appreciated by evaluating serial studies.

Complex regional pain syndrome (reflex sympathetic dystrophy)

- Complex regional pain syndrome, previously called reflex sympathetic dystrophy, causes persistent pain, tenderness, and swelling often due to minor trauma.
- Bone scan shows diffusely increased juxta-articular activity in multiple small joints of the hand or foot on delayed (skeletal) images. Blood pool and soft tissue phase uptake is variable, but most commonly both phases are increased.

Kidneys

Radiotracers

Tc-99m DTPA

- DTPA can measure glomerular filtration rate (GFR) and evaluate renal perfusion.
- DTPA is excreted by glomerular filtration. Approximately 20% is extracted by the glomerulus into the tubules with each pass. This is the identical extraction fraction as inulin, which is used to exactly measure GFR.

Tc-99m MAG3

- MAG3 can estimate renal plasma flow and evaluate renal perfusion, however MAG3 cannot measure GFR.
- MAG3 is filtered and excreted by the tubules. Greater than 50% is extracted by the glomerulus into the tubules with each pass. MAG3 is cleared predominantly by the proximal tubules with minimal filtration. It has a higher extraction fraction than DTPA, which provides better images in patients with renal insufficiency or obstruction.

Tc-99m DMSA

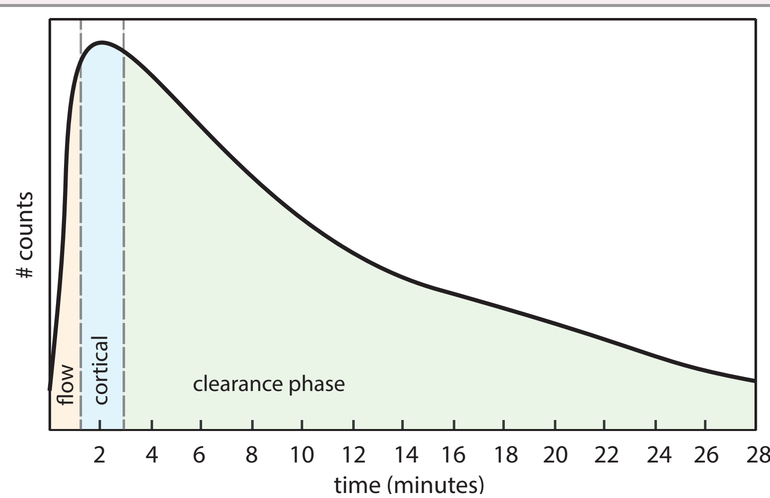
- DMSA scanning is performed in children to evaluate for the presence of renal scarring associated with pyelonephritis. The collecting system is not imaged.
- DMSA is a specialized cortical agent. It is bound in the renal tubules, which allows anatomic imaging of the cortex. It is the only renal tracer where SPECT imaging is performed.

Clinical applications of renal imaging

Renogram

- A nuclear renogram is a time-activity curve that provides a graphical representation of renal uptake and excretion. Approximately 10 mCi Tc-99m MAG3 (most commonly) or DTPA is administered. A normal renogram has three phases:

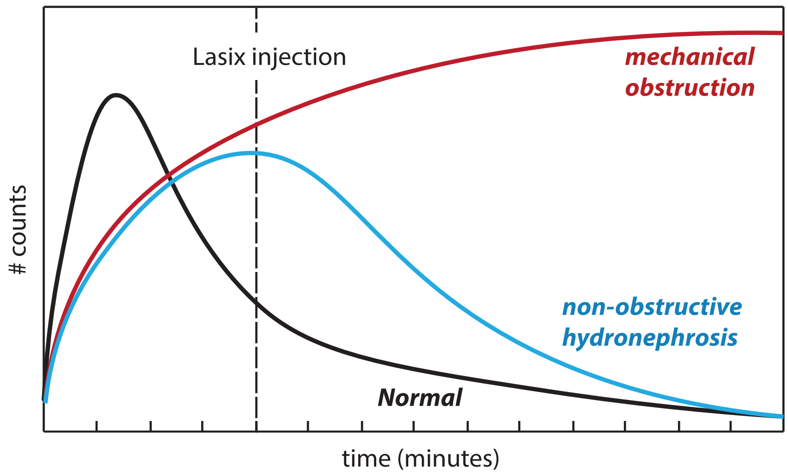
> 1) **Flow or perfusion phase** has a sharp upslope and lasts ~1 minute after injection. Normal kidneys are visualized within 2–5 seconds following visualization of the aorta. A slow upslope suggests decreased perfusion.
>
> 2) **Cortical function phase** usually occurs in the first 1–3 minutes, peaks at 3–5 minutes (= **time to peak activity**). Decreased renal function produces delayed cortical uptake.
>
> 3) **Clearance phase** is characterized by rapid urine excretion into the collecting system, which usually begins after 3 minutes. Normally half of the renal activity should be cleared in 8–12 minutes (= **half-time excretion**). Lack of pelvicalyceal clearance suggests obstruction.



- **Differential (or split) renal function** is determined during the cortical phase by counting the % activity in each kidney, which should be equal. Normal range is 45–55% for each kidney.
- Cortical transit of radiotracer can be quantified by the **20 minute-to-peak ratio**, which describes the percentage of maximal cortical activity remaining in each kidney at 20 minutes after injection. A ratio <0.3 is normal. Delayed transit reflects renal dysfunction.

Lasix renogram (diuretic renography)

- If hydronephrosis is present, a diuretic renogram can distinguish between obstruction and a nonobstructive cause of collecting system dilation. Both renal function and urodynamics are evaluated. The radiotracer is Tc-99m MAG3, much less commonly Tc-99m DTPA.
- After approximately 20 minutes of imaging, 40 mg IV **furosemide (Lasix)** is administered. A higher dose may be needed in patients with renal insufficiency. After infection, a fixed mechanical obstruction will show no change in the renogram curve. However, in a non-obstructive case, the additional pressure from diuretic effect will open up the collecting system and allow drainage of radiotracer from the kidney.



After administration of Lasix, a **clearance half-time** <10 minutes is normal.

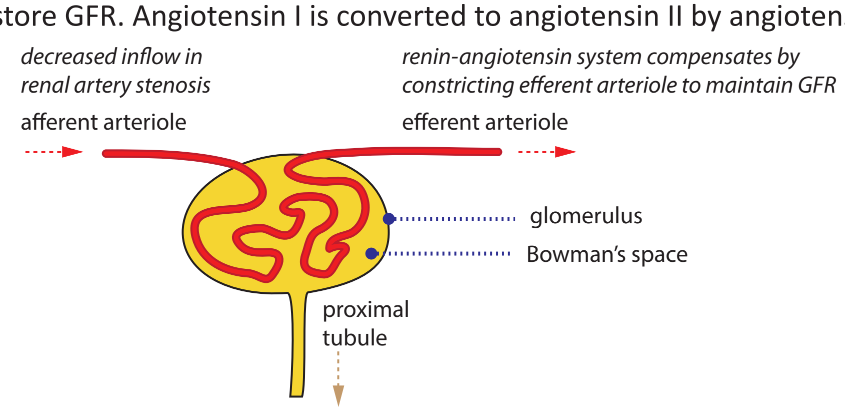
10–20 minutes is indeterminant.

>20 minutes suggests obstruction.

- False positives (Lasix renogram is positive for obstruction in the absence of a true obstruction) can be seen within dehydration, distended bladder, or renal failure (decreased response to diuretics).
- **Ethacrynic acid** is an alternative diuretic to Lasix in patients with sulfa allergies, though in practice Lasix is only associated with negligible risk of adverse reactions in these patients.

Angiotensin-converting enzyme (ACE) inhibitor renogram

- A positive ACE renogram is relatively specific for renal artery stenosis, which may be a cause of hypertension. The radiotracer is Tc-99m MAG3, much less commonly Tc-99m DTPA.
- In compensated renal artery stenosis, decreased blood to the glomerulus stimulates renin → angiotensin I → angiotensin II. Angiotensin II constricts the efferent (outgoing) arterioles to restore GFR. Angiotensin I is converted to angiotensin II by angiotensin converting enzyme.



ACE inhibitor renogram (continued)

- After administration of an ACE inhibitor (typically captopril, 25–50 mg), the efferent arterioles will relax and GFR will decrease.

  If a patient is already on an ACE inhibitor, it should be stopped for 48 hours to 1 week prior to the exam.

  Intravenous access should be maintained in case fluid resuscitation is needed to treat hypotension.

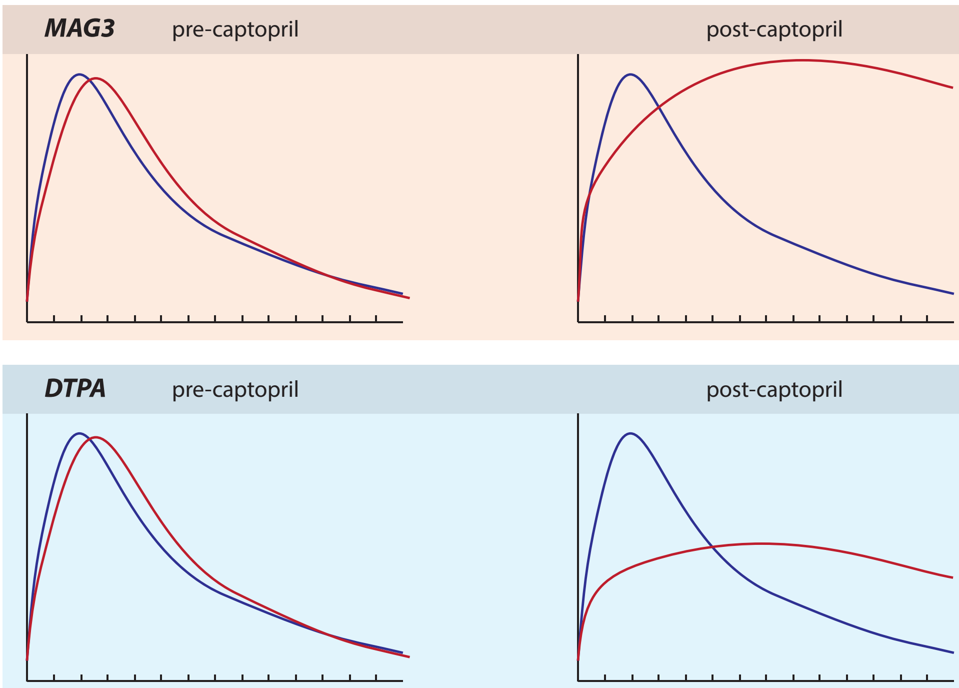
  Known severe renal artery stenosis is a relative contraindication as ACE inhibitor-induced intrarenal hypotension can lead to acute renal failure.

- An ACE renogram can be performed in one or two days.

  A **single-day, two-stage protocol** comprises a baseline non-captopril study using a low dose of MAG3, followed by a captopril study several hours later when residual activity from the first study has cleared the kidneys and urinary tract.

  A **two-day protocol** is usually used for patients without pre-existing renal dysfunction. The captopril study is performed first, which if normal, excludes renovascular hypertension and thus eliminates the need for a baseline study. If the captopril study is abnormal, a baseline study is performed on the following day.

- A positive study depends on the radiotracer utilized (MAG3 or DTPA), but the hallmark is a post-ACE inhibitor renogram that becomes abnormal or more abnormal, usually unilaterally.



  MAG3 (tubular agent): Uptake and secretion are generally preserved even with a reduced GFR, but further ↓ GFR results in ↓ urine production and ↓ washout of secreted agent from collecting system.

  Positive criteria for MAG3 study include: <40% uptake by one kidney at 2–3 minutes; difference in cortical activity by 20%; or delay in time to peak activity of more than 2 minutes (compared to pre-ACE inhibitor).

  DTPA (glomerular agent): Unlike MAG3, ↓ GFR causes diminished uptake and excretion of DTPA.

- Bilateral worsening renograms can be caused by hypotension, dehydration (most common), urinary obstruction, medical renal disease, or bilateral renal artery stenosis.

Renal cortical imaging

- Renal cortical imaging is limited to renal imaging in children to evaluate for pyelonephritis or scarring. The radiotracer is Tc-99m DMSA.
- A normal DMSA study excludes the diagnosis of acute pyelonephritis.
- A positive study for pyelonephritis can have three patterns:

  Focal cortical defect.

  Multifocal cortical defects.

  Diffusely decreased radiotracer.

Radionuclide cystography (RNC)

- Radionuclide cystography is the most sensitive test for evaluation of reflux in pediatric population with less radiation exposure compared to voiding cystourethrography (VCUG).
- The radiotracer chosen is variable. Tc-99m pertechnetate, Tc-99m DTPA, and Tc-99m sulfur colloid can all work well. Regardless of the radiotracer chosen, it is injected retrograde into the catheterized bladder.
- Grading of reflux:

  Minimal (RNC grade I): Reflux is confined to the ureter.

  Moderate (RNC grade II): Reflux extends superiorly to the pelvicalyceal system.

  Severe (RNC grade III): Severe reflux causing a tortuous ureter and/or dilated intrarenal system.

Renal transplant evaluation

- While transplanted kidneys are often initially evaluated by ultrasound and ultimately by biopsy, radionuclide renogram using Tc-99m MAG3 or DTPA is an important diagnostic tool.
- A normal renal transplant should be visualized on the flow phase within seconds after the iliac vessels. Its cortical and clearance phases are similar to those of a native kidney.
- Renogram findings of common transplant complications include:

Acute tubular necrosis (ATN) occurs within 3–4 days after surgery and results from perioperative ischemia. Its renogram is characterized by normal perfusion, but poor renal function with increased cortical retention, delayed clearance and decreased urine excretion.

Hyperacute rejection produces immediate vascular thrombosis in the transplanted kidney, resulting in absent perfusion, markedly reduced or absent function.

Acute rejection occurs within weeks to three months after surgery. It presents as decreased perfusion and marked cortical retention.

Chronic rejection occurs months to years after surgery. It is characterized by progressive worsening perfusion and function with cortical thinning, pelvicalyceal dilation.

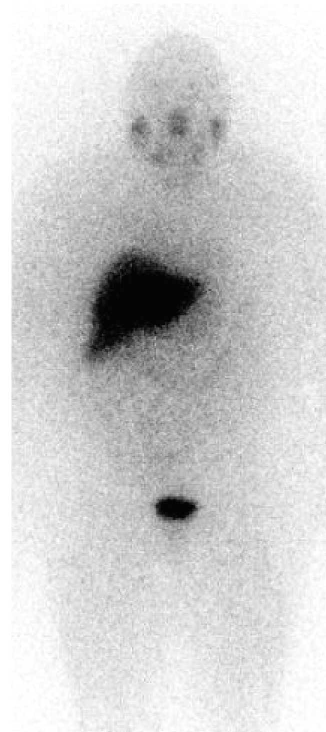
Cyclosporin nephrotoxicity appears similar to ATN, but occurs at a later time after ATN should have resolved.

Urine leak presents as progressive accumulation of activity outside the urinary tract.

Urinoma, **hematoma** and **lymphocele** may present as fixed perirenal photopenic areas.

Radiotracers

I-123 MIBG



Planar image shows normal distribution of I-123 MIBG.

- I-123 MIBG is a norepinephrine analog that is taken up by chromaffin cells of sympathetic adrenergic tissue. It is used to image pheochromocytoma in adults and neuroblastoma in children. To a lesser extent, MIBG is taken up by other neuroendocrine tumors including carcinoid, medullary thyroid carcinoma, and paraganglioma.
- I-131 MIBG can be used to treat neuroblastoma in children.
- Normal distribution is areas of sympathetic innervation, including salivary glands, heart, thyroid (although this is typically intentionally blocked with Lugol's solution to reduce thyroid dosimetry), liver, kidney, and bladder.

Indium-111 pentetreotide (Octreoscan)

- Indium-111 is cyclotron-produced and decays by electron capture, emitting two gamma photons at 173 and 247 keV. Its half-life is 67 hours. Pentetreotide is octreotide conjugated to DTPA which allows binding to In-111.
- In-111 pentetreotide is used to detect tumors with somatostatin receptors, including amine precursor uptake and decarboxylation (APUD) tumors. Most common applications are evaluation of carcinoid or islet cell tumors such as gastrinoma. However, pentetreotide has low sensitivity for insulinoma.
- Paragangliomas (extra-adrenal pheochromocytomas) are seen better with In-111 pentetreotide compared to MIBG, although In-111 pentetreotide is uncommonly used for detection of paragangliomas.
- Normal distribution is intense renal and splenic uptake, with slightly less hepatic uptake.

Indium-111 oxine leukocytes (WBCs)

- The normal distribution of Indium-111 labeled WBCs is spleen > liver >> bone marrow.
- Infection imaging with an In-111 WBC scan is performed at 24 hours.
- The key advantage of In-111 oxine WBCs compared to gallium is the lack of physiologic bowel accumulation, which allows evaluation of abdominal or bowel infection/ inflammation.
- Disadvantages compared to gallium include a tedious labeling procedure, higher radiation dose, and less accuracy in diagnosing spinal osteomyelitis.
- Advantages of In-111 WBC scan compared to Tc-99m HMPAO include absence of interfering bowel and renal activity, ability to perform delayed imaging, and ability to perform simultaneous Tc-99m sulfur colloid or Tc-99m MDP bone scan. These combined approaches are very helpful for evaluation of osteomyelitis in the setting of a baseline abnormal bone scan (e.g., prosthesis evaluation), discussed below.

Tc-99m HMPAO leukocytes (WBCs)

- The advantages of a WBC scan labeled with Tc-99m HMPAO compared to In-111 are related to the shorter half-life of Tc-99m, which allows a higher administered activity, better counts, a lower absorbed dose, and ability to perform earlier imaging. For these reasons, Tc-99m is often preferred in children in appropriate cases.
- However, a major disadvantage is physiologic uptake within the gastrointestinal and genitourinary tracts due to unbound Tc-99m HMPAO complexes, which limits bowel evaluation. Renal activity occurs early, while bowel activity is seen after 1–2 hours. Additionally, delayed imaging is less practical with Tc-99m due to its shorter half-life.

Gallium-67

- Gallium-67 is cyclotron-produced and decays by electron capture, with a half-life of 78 hours. It emits multiple gamma rays at 93, 184, 296, and 388 keV (easier to remember as 90, 190, 290, and 390 keV). Gallium binds to transferrin, which is found in infection, inflammation, and neoplasm.
- Normal distribution includes the nasopharynx, lacrimal and salivary glands, liver, colon, and bone marrow. Use is limited in the abdomen due to high bowel and liver uptake.
- Persistent gallium in the kidneys is never normal after 24 hours and signifies renal disease or obstruction.
- Diffuse pulmonary uptake is also not normal, with a wide differential of infectious and inflammatory conditions:

  Pneumocystis pneumonia, idiopathic pulmonary fibrosis, sarcoidosis, lymphangitic carcinomatosis, miliary tuberculosis, and fungal infection.
- The *panda sign* reflects increased uptake in the nasopharynx, parotid glands, and lacrimal glands due to inflammation, resembling the dark markings on a panda's face. It is classically seen in sarcoidosis.

  The *panda sign* is not specific for sarcoidosis, although it strongly suggests sarcoid when the *λ sign* is also seen (due to bilateral hilar and right paratracheal adenopathy). Sarcoid is also the most common disorder to affect the salivary and lacrimal glands symmetrically.

  Differential diagnosis of the *panda sign* includes Sjögren syndrome, lymphoma after irradiation, and AIDS.

Normal Gallium-67 scan (5.25 mCi administered) shows high uptake in the bowel and salivary glands, moderate uptake in the liver and bone marrow, and faint uptake in the lungs.

- Gallium scans have been largely replaced by F-18 FDG PET/CT. A residual first-line application of gallium scan is the evaluation of spinal discovertebral osteomyelitis.

Thallium-201

- Thallium-201 is cyclotron-produced, decays by electron capture, and produces relatively low-energy characteristic x-rays of 69–81 keV. Its half-life is 73 hours. Normal distribution is prominent uptake in the kidneys, heart, liver, thyroid, and bowel.
- Thallium imaging is infrequently used due to its long half-life and resultant high radiation exposure. Historically, thallium has been used in combination with gallium scan to distinguish between Kaposi sarcoma, lymphoma, and tuberculosis in immunocompromised patients. Thallium was also historically used as a myocardial agent.

Clinical applications

Hepatocellular carcinoma (HCC)

- Historically, increased focal gallium uptake in the liver suggests hepatocellular carcinoma (HCC). Conversely, HCC is extremely unlikely if gallium uptake is diminished.

Combined gallium and thallium imaging

- Kaposi sarcoma is thallium-avid, but does not take up gallium. *Mnemonic: KaT (Kaposi is Thallium-avid).*
- Tuberculosis and atypical mycobacteria take up gallium but not thallium. This is the opposite of Kaposi sarcoma. *Mnemonic: TuG (Tuberculosis is Gallium-avid).*
- Lymphoma takes up both gallium and thallium. *Mnemonic: Lymphoma likes both.*

Osteomyelitis

- A positive triple-phase Tc-99m MDP bone scan is only specific for osteomyelitis if the radiograph is normal. If there is an underlying abnormality, SPECT/CT, gallium or WBC scan can increase specificity.
- Gallium imaging can increase specificity of a positive bone scan, especially for vertebral osteomyelitis and discitis.
- A WBC scan (typically with Indium-111) can increase specificity in evaluation of an infected orthopedic prosthesis, where scanning is typically performed in conjunction with Tc-99m sulfur colloid. When comparing the two scans (WBC and sulfur colloid), a discordant region with increased WBC tracer uptake and decreased sulfur colloid uptake is suggestive of marrow replacement by WBCs, consistent with osteomyelitis. Conversely, a region of normal marrow would be expected to demonstrate uptake both by WBCs and sulfur colloid.

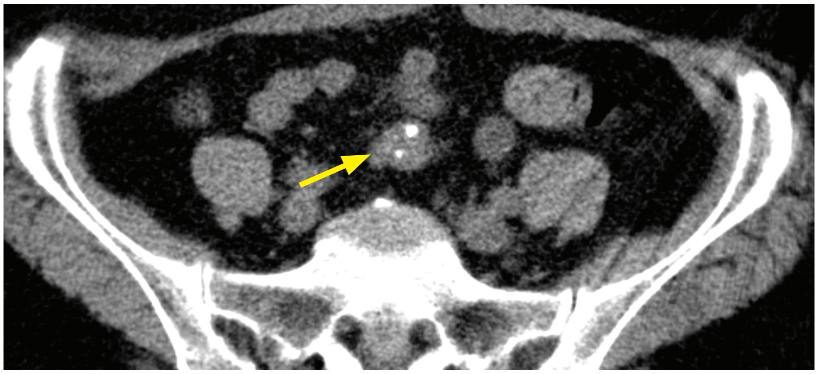
Adrenal pheochromocytoma

- I-123 MIBG is generally considered the first-line agent for evaluation of adrenal pheochromocytoma. Indium-111 pentetreotide can be considered as a second-line agent if the MIBG scan is negative, though the normal renal uptake of pentetreotide may obscure an adrenal lesion.

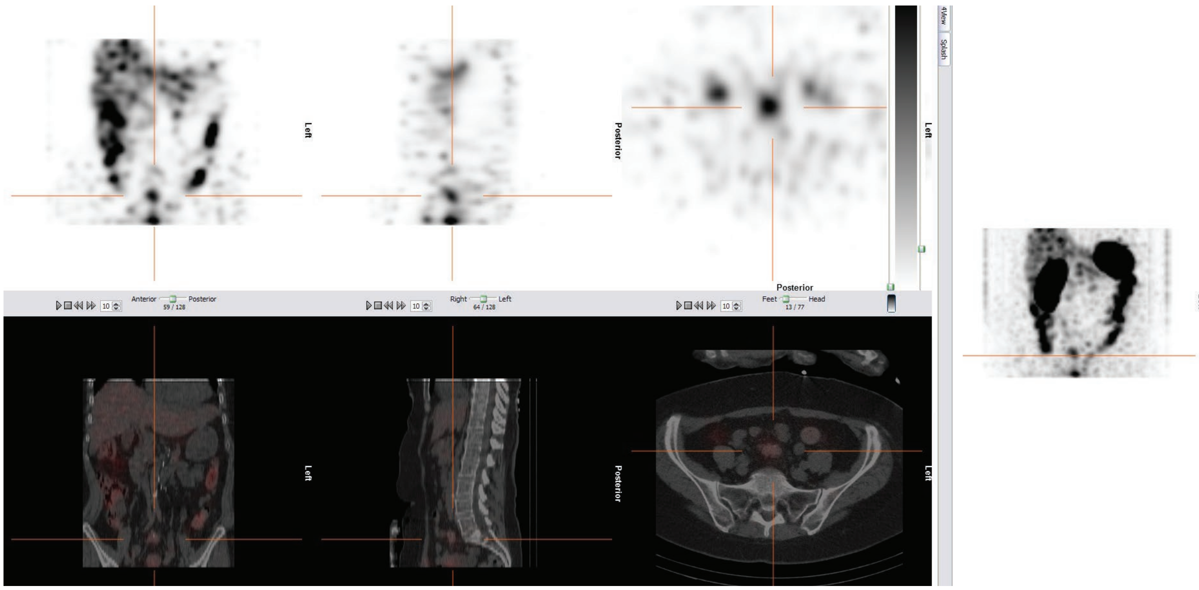
Extra-adrenal pheochromocytoma

- The sensitivity of I-123 MIBG for detection of extra-adrenal pheochromocytoma (i.e., paraganglioma or glomus tumor) ranges from 63–100%, with potential loss of MIBG uptake seen in tumor cell dedifferentiation, altered membrane transport proteins, or interference by medications. Indium-111 pentetreotide is also an option.
- The role of F-18 FDG PET in the evaluation of metastatic extra-adrenal pheochromocytoma is evolving. Most tumors are FDG-avid.

Metastatic carcinoid/neuroendocrine tumor



Metastatic carcinoid: Axial non-contrast CT shows a mesenteric mass containing coarse calcification (arrow) in the central lower abdomen, which has a typical appearance for carcinoid.



Fused multiplanar SPECT (top row) from an indium-111 pentetreotide scan and CT (bottom row) correlate a focus of increased uptake in the lower abdomen with the calcified mesenteric mass on CT, confirming the diagnosis of neuroendocrine tumor/carcinoid. All other foci of tracer uptake seen on SPECT localized to bowel, representing physiologic bowel uptake.

- Indium-111 pentetreotide (Octreoscan) was traditionally the tracer of choice for evaluation of carcinoid tumor (see below). SPECT/CT is almost always performed in conjuction with an Octreoscan.
- The sensitivity for detecting neuroendocrine tumors is 82–95%; however, specificity is only around 50%, especially when a “hot spot” is found near a region of physiologic uptake in the pituitary, thyroid, liver, spleen, urinary tract, or bowel. False positives can also occur in benign processes such as inflammation, Graves disease, and sarcoidosis. These inflammatory false-positives are thought to be due to somatostatin receptors expressed in activated lymphocytes.
- Octreoscan is being replaced by Gallium-68 DOTATATE PET/CT (discussed earlier in this chapter) for detection of neuroendocrine tumors with high somatostatin receptor expression, as PET/CT has higher spatial resolution than SPECT/CT and Gallium-68-DOTATATE has much higher affinity for somatostatin receptors than Indium-111 pentetreotide.

Other non-PET imaging

Lymphoscintigraphy

- For cancers that metastasize first to regional lymph nodes, such as melanoma and breast carcinoma, lymphoscintigraphy can be used to identify the sentinel lymph node required for tumor staging. Filtered Tc-99m sulfur colloid is injected in the peritumoral region, subsequent images allow mapping of lymphatic drainage from the tumor. The first node visualized is the sentinel node. A gamma probe can be used intraoperatively during the same-day surgery to localize the sentinel node for excision.
- Lymphoscintigraphy is also utilized in the diagnosis of lymphedema. Lymphatic drainage in the involved extremity is visualized after interdigital injection of the radiotracer. Obstructive lymphedema presents as delayed migration of activity from the injection site, diffuse dermal backflow uptake, dilated and collateral lymphatic channels.

Molecular breast imaging

- Technetium-99m sestamibi concentrates in breast cancers and can be used as a problem-solving adjunct to mammography and ultrasound, particularly in women with dense breast tissue.
- Normal breast has low, homogeneous or patchy uptake. Focal uptake in the breast or axilla is suspicious for malignancy.
- Molecular breast imaging with Tc-99m sestamibi has high sensitivity and specificity (80–90%). However, radiation dose to the breast remains a disadvantage.

Radionuclide testicular imaging

- Historically, testicular imaging with technetium-99m pertechnetate has been used in the diagnosis of testicular torsion. The radiotracer is injected intravenously, immediate dynamic imaging of the anterior pelvis is then obtained, followed by delayed static images.
- Normal testes show symmetric homogeneous uptake. Decreased perfusion or focal photopenia on the symptomatic side is diagnostic of torsion. Increased perfusion suggests hyperemia, which is seen in epididymitis/orchitis.
- Although radionuclide imaging is more sensitive for testicular torsion, it is slower and less readily available compared to scrotal ultrasound.